The European Open Web Index — Current State and Perspective

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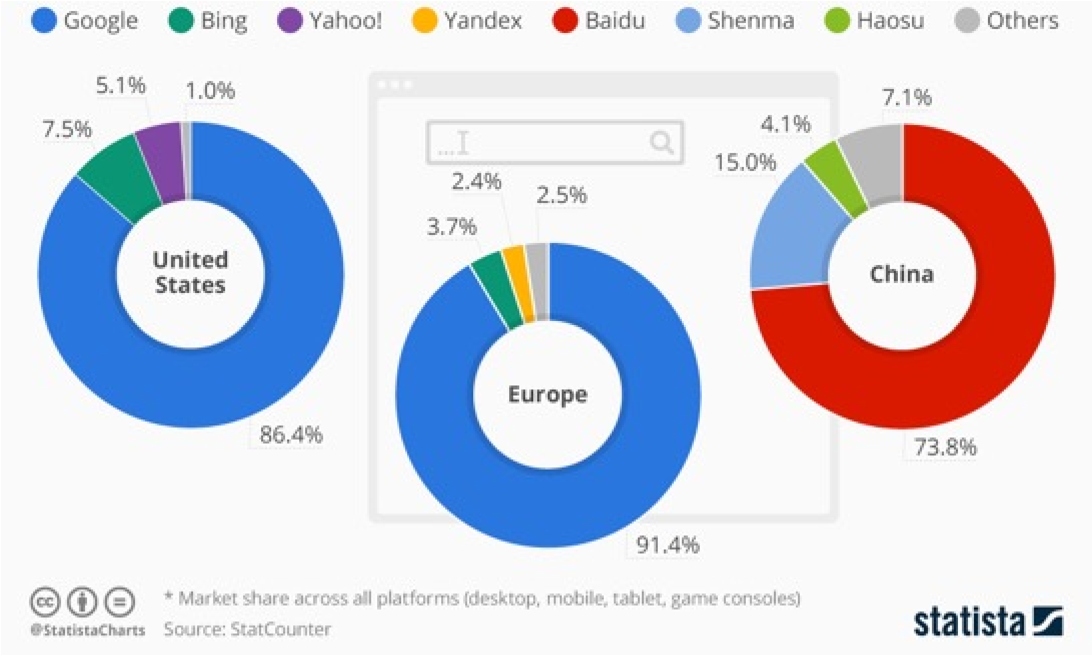
Web Search Today

A critical resource managed as an oligopoly

Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



But does it matter? Yes — on four dimensions

User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

> Being able to navigate the Web is as fundamental as GPS or the power grid

Challenges

THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

VOLUME

1.15 Billion

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

Top 0.1%–1% of sites

may account for roughly half of all pages

heavy-tail intuition from large web graph studies

~ 400 Billion

documents in Google’s search index

2020 figure cited in US v. Google testimony

Sites ≠ domains ≠ pages; use exact counts cautiously

VELOCITY

+8.6 M / month

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

~ 3.2 / second

same monthly net change, averaged across December

149 ZB (global)

all digital data created / copied worldwide in 2024

IDC / Statista global datasphere estimate

The web changes constantly; crawling is continuous, not one-off

VARIETY

150+ languages

observed across websites

W3Techs usage survey

249 CMS Platforms

CMS platforms identified in the HTTP Archive dataset

HTTP Archive Web Almanac 2024 (Wappalyzer-based)

HTML · PDF · JSON-LD · media · APIs

content types to parse and process

HTML · PDFs · JS apps · schemas · media · APIs

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) · US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista

- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

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- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
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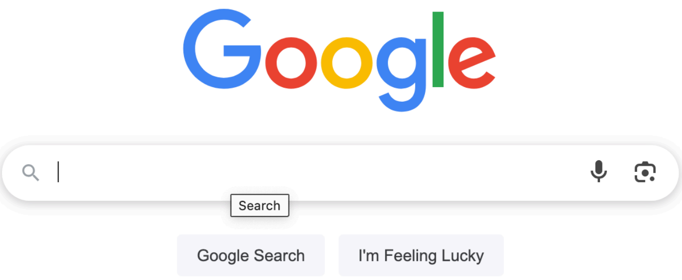
High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

Reduced user choice

Single source of our information: We would not be happy with only one European newspaper



Newspaper map: reddit.com/r/MapPorn/comments/1c70v2q/newspapers_of_record_or_note_in_europe/

Search Engine Manipulation Effect

- Users locked into a single information gatekeeper
- **Search Engine Manipulation Effect (SEME)**: biased search rankings can shift voting preferences of undecided voters by **20% or more** — up to **72%** in some demographics — while users remain largely unaware of the manipulation

Epstein, R. & Robertson, R.E. "The search engine manipulation effect (SEME) and its possible impact on the outcomes of elections." PNAS 112(33), E4512–E4521, 2015. DOI: 10.1073/pnas.1419828112

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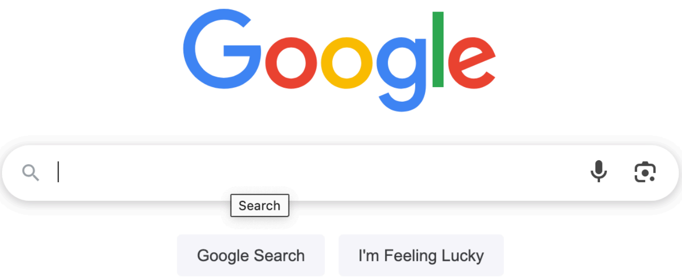

vs.

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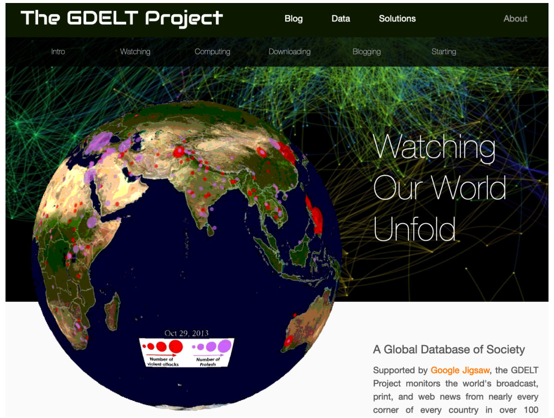
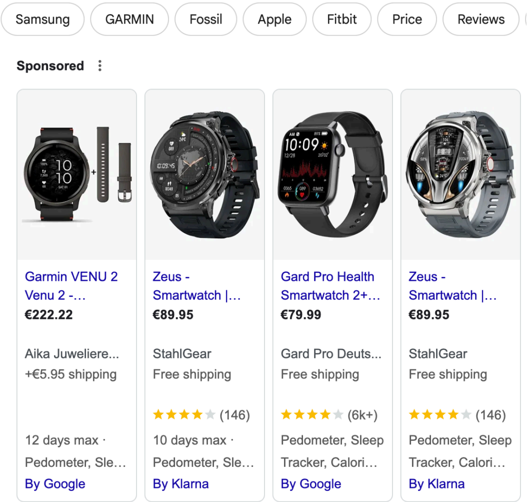
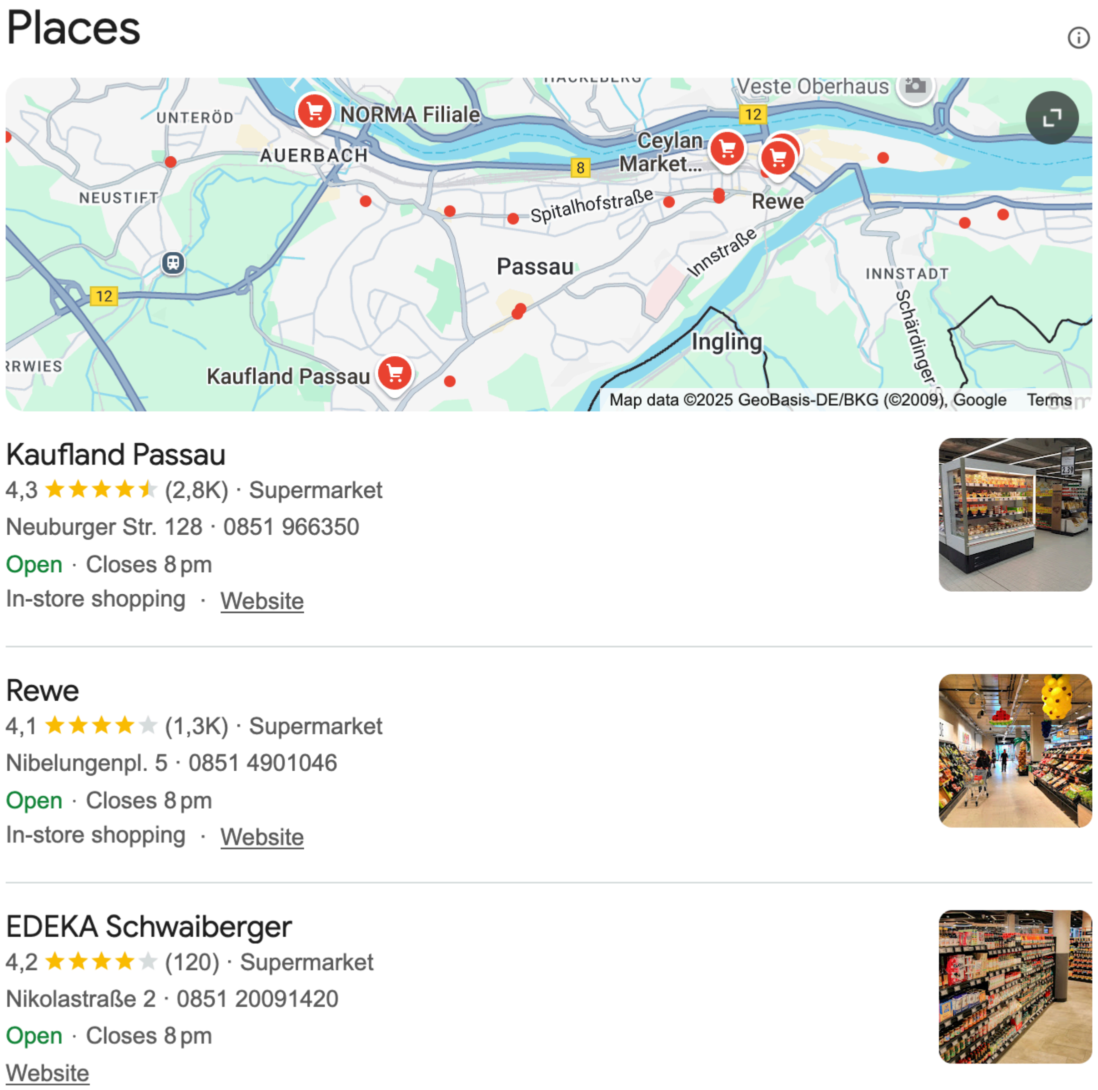
The web is the richest information source

The Web is the richest information source, but we are blind on what is there

- Companies, products, scientific information, events, geo-location data
- 60% of web pages contain structured data in form of Microdata / Microformats
  - Products, FAQ, Places, Shops
  - Job postings, Reviews, Events







Societal Challenges which require a grip on the Web

Misinformation / Disinformation

- Monopolistic ranking creates single points of failure for information quality
- Lack of index transparency prevents independent auditing and cross-language disinfo detection
- Open indices enable reproducible fact-checking pipelines

Guidance & Algorithmic Bias

- Search autocompletions and knowledge panels shape health, legal and consumer decisions
- Filter bubbles create asymmetric information through personalization
- Low-income users disproportionately affected by low-quality results

Defence & Security

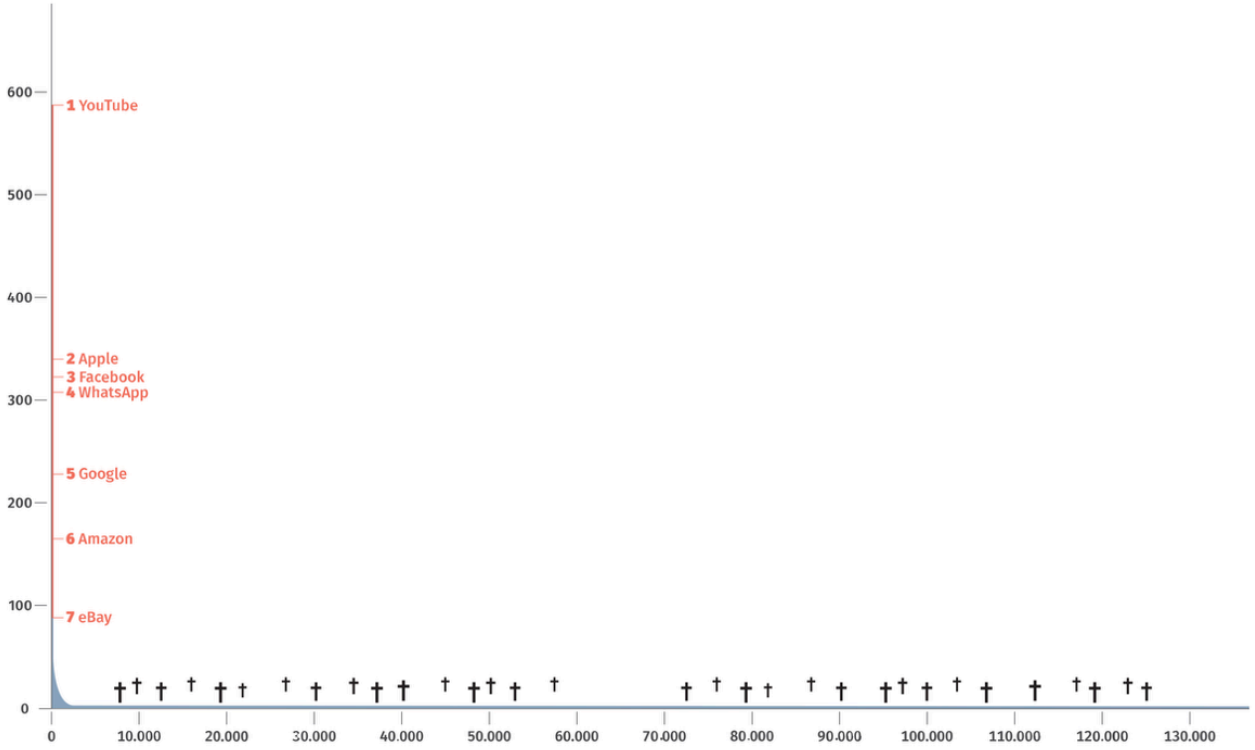
- European OSINT depends on US-controlled indices — strategic vulnerability in crisis scenarios
- Adversarial SEO ("search engine poisoning") is a documented hybrid warfare tactic
- No sovereign search infrastructure for independent web monitoring

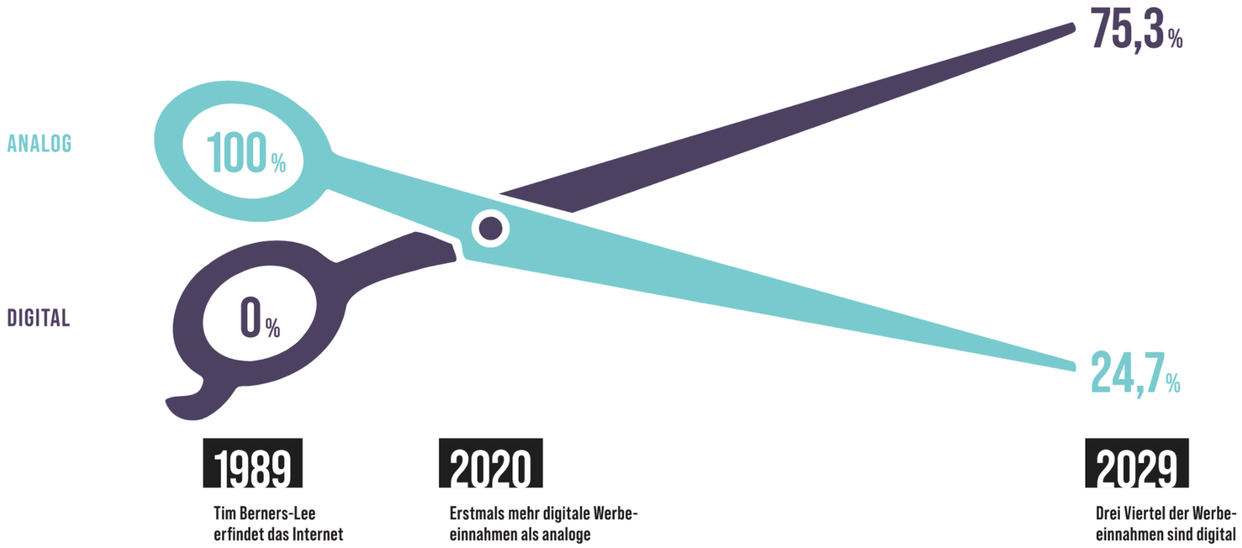
Digital & Information Literacy

- Most users cannot distinguish ads from organic results — "trust by ranking" effect
- Algorithmic literacy remains low even among university students
- Transparent, inspectable search infrastructure is a prerequisite for teaching information literacy effectively

Europe is not present in the web

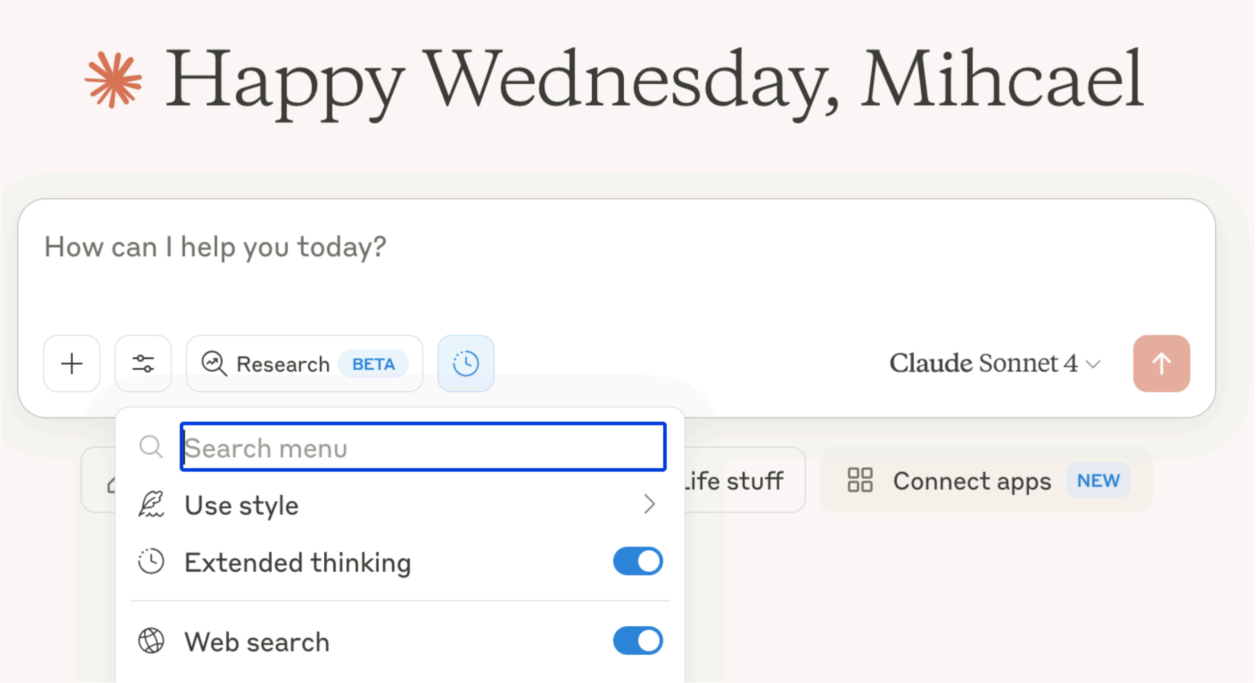
- Advertising revenue consolidated by US gatekeepers
- European content, languages, and perspectives underrepresented in search results
- No European alternative for web-scale data access

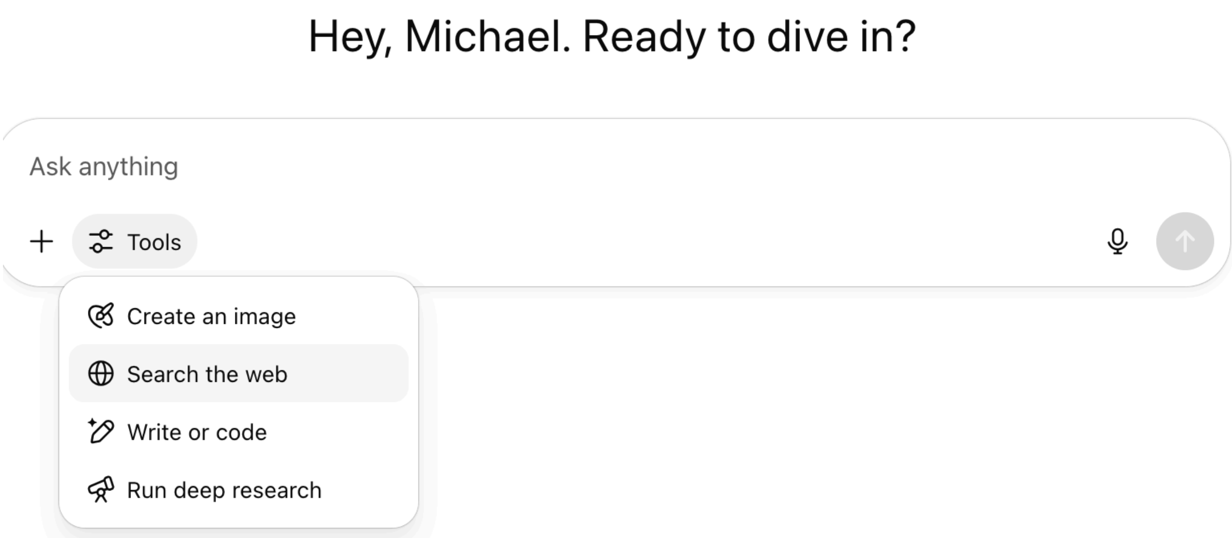




The web drives innovation and AI

- Web data is essential for training LLMs and VLMs
- Retrieval Augmented Generation (RAG) requires web-scale document collections
- Agentic AI systems rely on web search as a core tool
- Web search remains an integral part of modern AI pipelines





OpenWebSearch.eu — Our Mission

> Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

Four Objectives

1. **Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
2. **Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
3. **Added Value Services** Dashboard, tools, data access APIs
4. **Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

14 Core Partners

Infrastructure

Research

NGOs

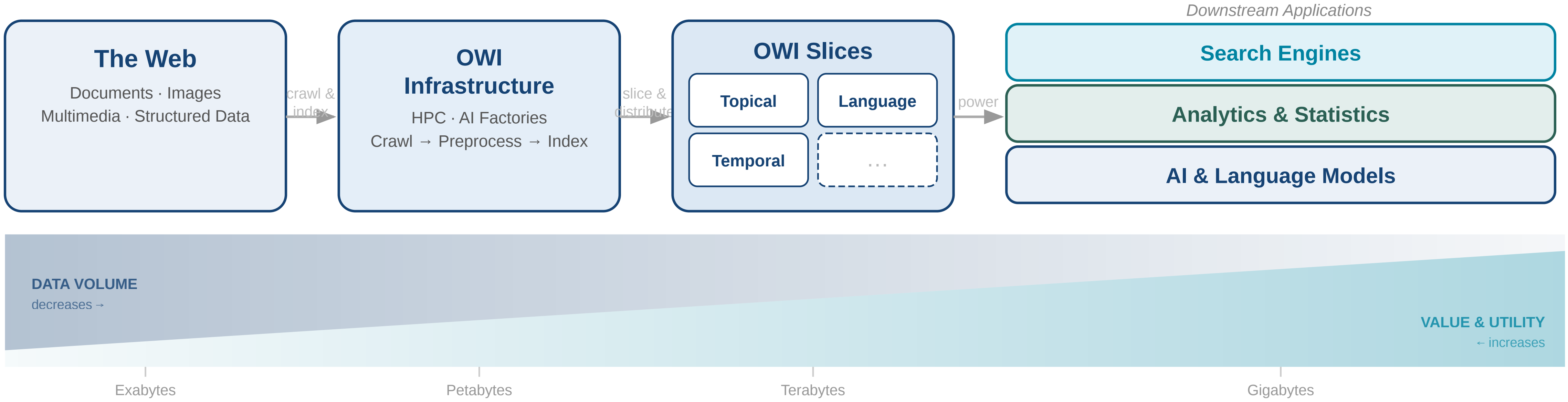
Businesses

Plus 16 financially supported third-party projects (FSTPss)

The Open Web Index (OWI)

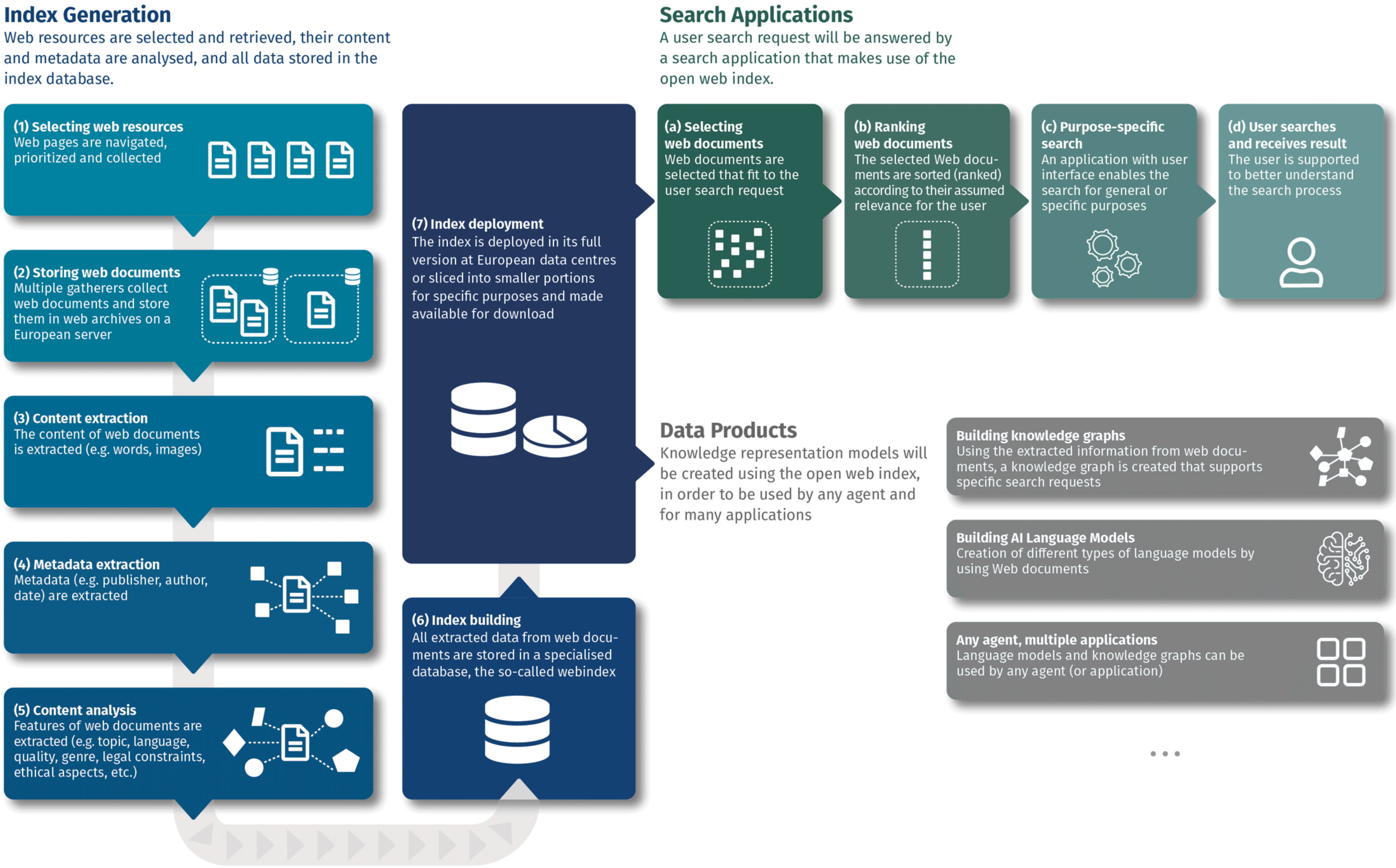
A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

Our Proposition: A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe

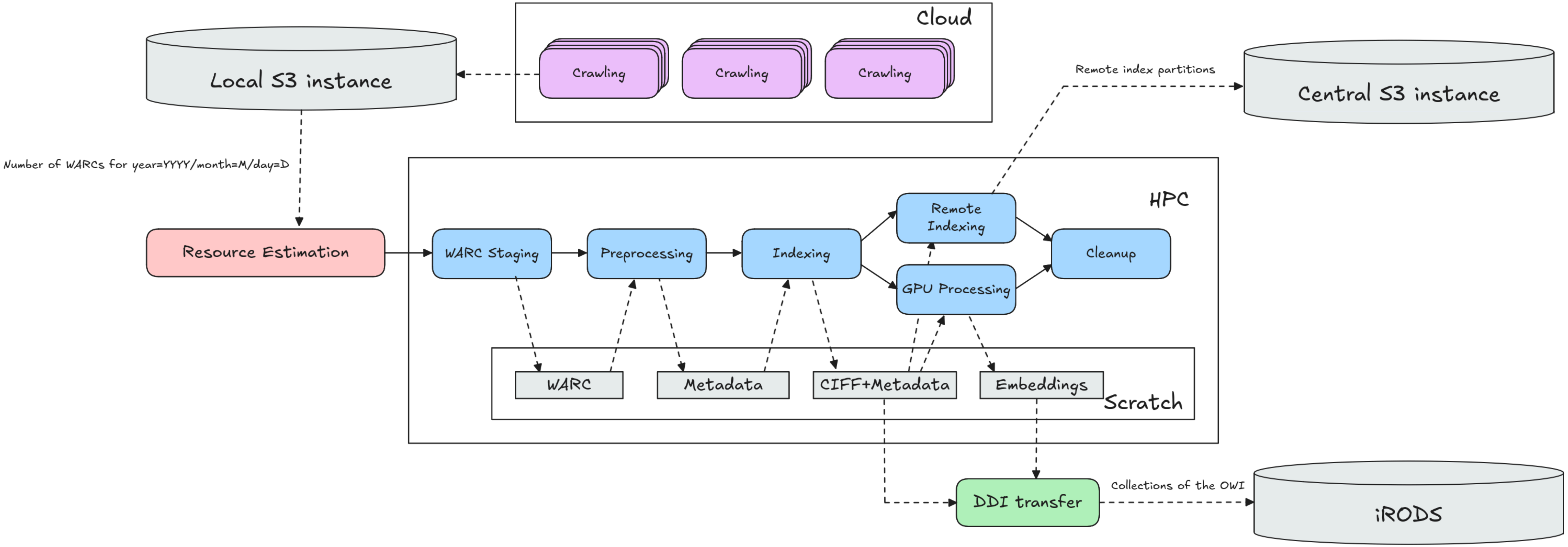


Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." Journal of the Association for Information Science and Technology (2023).

Conceptual Workflow



Daily Workflow Pipeline



Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

The OpenWebSearch.eu Pipeline — Overview

Step 1 — Crawling

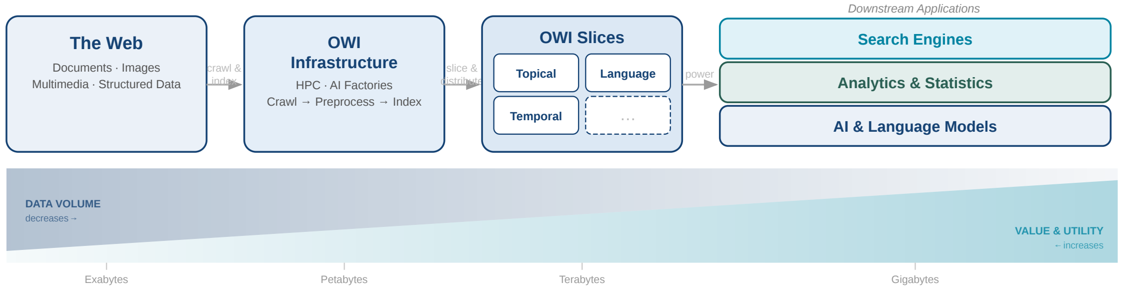
- 100 M URLs / day, 25 machines in 3–4 DCs
- European domain focus
- Respects robots.txt, Gen AI opt-out
- Full transparency via Dashboard

Step 2 — Preprocessing (Resilipipe)

- HTML → clean text (ratio 1:5)
- Metadata: JSON-LD, Microdata, OpenGraph
- Semantic enrichment: language, topics, NER, geo-locations
- Structured data from 60 % of pages

Step 3 — Index & Distribution

- Sparse full-text index (CIFF / Lucene)
- Feature index: link structure, style, genre
- Dense index: Jina V3 embeddings
- 810+ daily datasets via Owilix / Dashboard



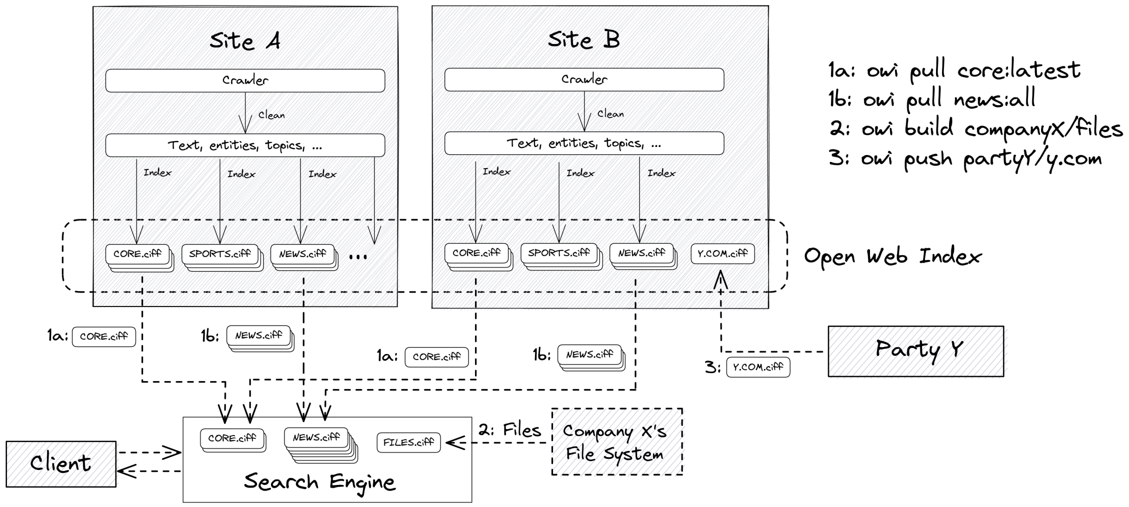
Scale (as of early 2026)

- **25 VMs** across 3–5 data centres (IT4I, CSC, LRZ, CERN) running 24/7
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index in **810+ public datasets**
- Factor **20 cheaper** than commercial indices

Architecture principles

- Federated storage (iRODS + S3), cross-DC workflow via LEXIS
- Client-side SQL access via Owilix CLI
- Open format: CIFF sparse index + Parquet files
- Dense embeddings (Jina V3) available on select slices

Architecture & Scale



Four Cluster Tiers per Data Center

- **CCT** — OWLer crawler (StormCrawler-based); produces WARC files into S3
- **CFT** — singleton ScyllaDB URL frontier at CSC; coordinates all crawl sites
- **PET** — Resilipipe (Python multiprocessing + Resiliparse): WARC → Parquet
- **IST** — Open Web Indexer (Apache Spark): Parquet → CIFF + remote index

Deployed at: IT4I@VSB · BADW-LRZ · CSC · CERN + FSTP partners

Scale (early 2026)

- **25 VMs** across 3–5 data centers running 24/7
- **100 M URLs / day** crawling · **~1.1 PB** raw data
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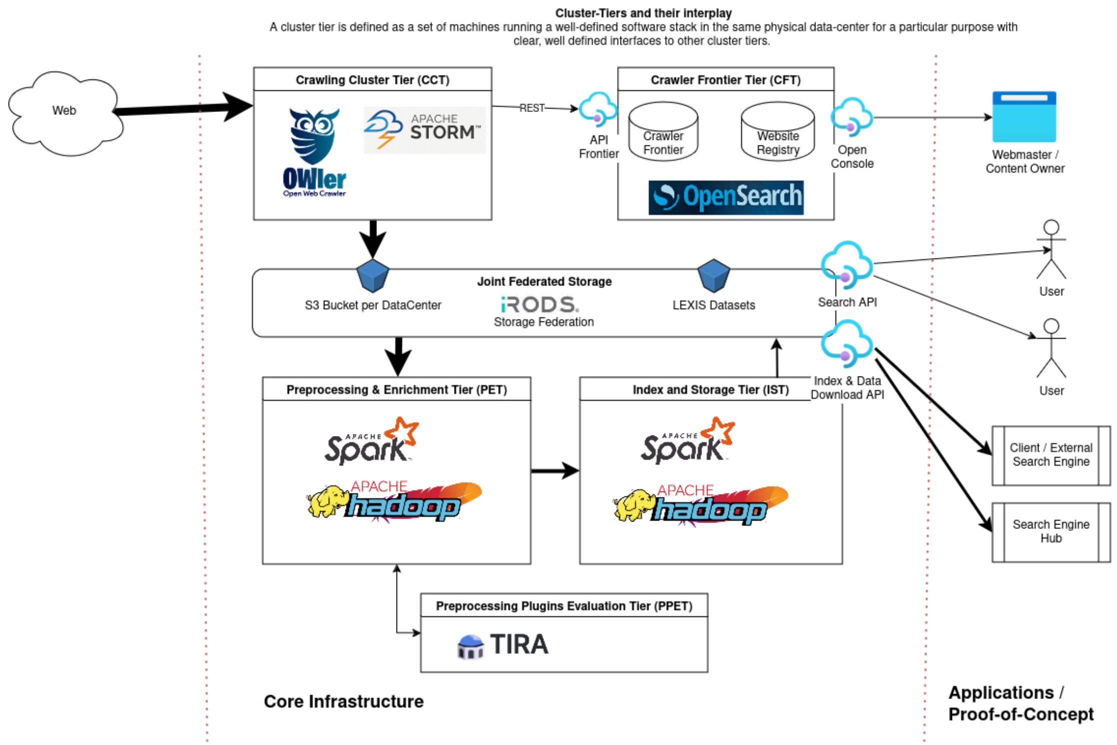
Cluster Tiers in Detail



Crawling (WP 1)

- **OWLer** (distributed StormCrawler): multi-site web crawl, WARC output
- Peak: ~100 M URLs/day at 4 partner sites (UNI PASSAU, LRZ, IT4I, CSC + EXAION)
- **ScyllaDB** frontier: dedup, politeness, cross-DC coordination

Preprocessing (WP 2)

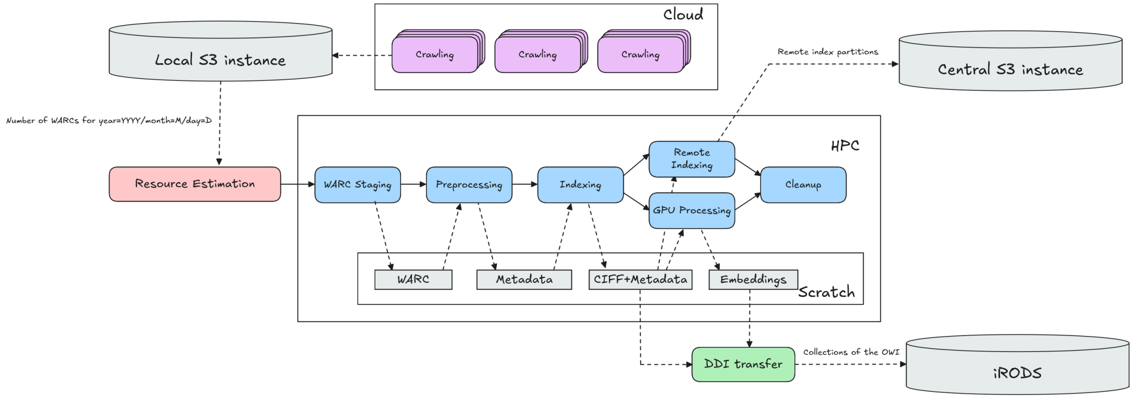
- **Resilipipe** on Spark: HTML → clean text, metadata extraction, topic/entity tagging
- **PPET** (singleton): plugin evaluation via TIRA before promotion to production
- GPU tier: quality-filtered docs → Jina V3 dense embeddings

Indexing (WP 3)

- **Open Web Indexer** on Spark: Parquet → CIFF shards per collection + language

- Also produces **remote index** (DuckDB/OUR²S-queryable Parquet partitions)
- Federated storage: **iRODS** + S3 via LEXIS DDI layer

Daily Workflow

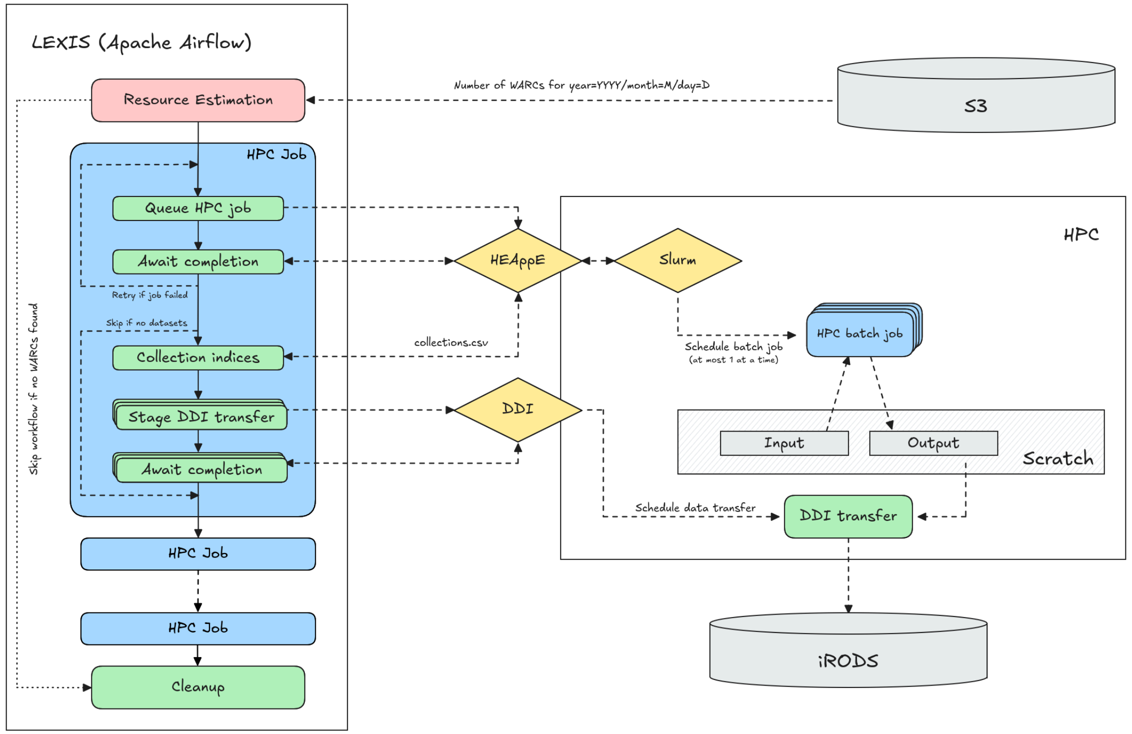


Each day at **03:00 AM** the pipeline triggers:

1. **WARC Staging** — copy new WARC files from S3 to HPC scratch
2. **Resilipipe** — WARC → gzipped Parquet (text, metadata, enrichments)
3. **Open Web Indexer** — Parquet → CIFF shards per collection / language
4. **GPU Processing** — Resilipipe GPU → Jina V3 embeddings for quality slice
5. **Remote Index** — Parquet partitions for DuckDB / OUR²S remote queries
6. **DDI Transfer** — push output collections to iRODS via LEXIS DDI layer
7. **Cleanup** — remove intermediate HPC scratch data

All intermediate data stored on **HPC scratch** between steps — restartable on partial failure.

Workflow Orchestration via LEXIS



LEXIS (Apache Airflow) manages the full daily pipeline:

- **Resource estimation**: counts WARCs for that date → computes Slurm allocation
- **HEAppE**: submits and monitors HPC batch jobs at each data center
- **DDI layer**: transfers output collections to iRODS after each job succeeds
- Automatic retry on failure; session tokens refreshed throughout

Workflow configuration (JSON per DC):

- `crawl_location` / `hpc_location` / `storage_location` — Airflow Connections
- Per-task: `resources`, `max_retries`, `stage_ddi`, `depends_on`
- Dev / test / prod variants with reduced resource profiles

Data center connections: S3 (crawl), HEAppE (HPC scheduling), iRODS (persistent storage) — fully configurable per data center.

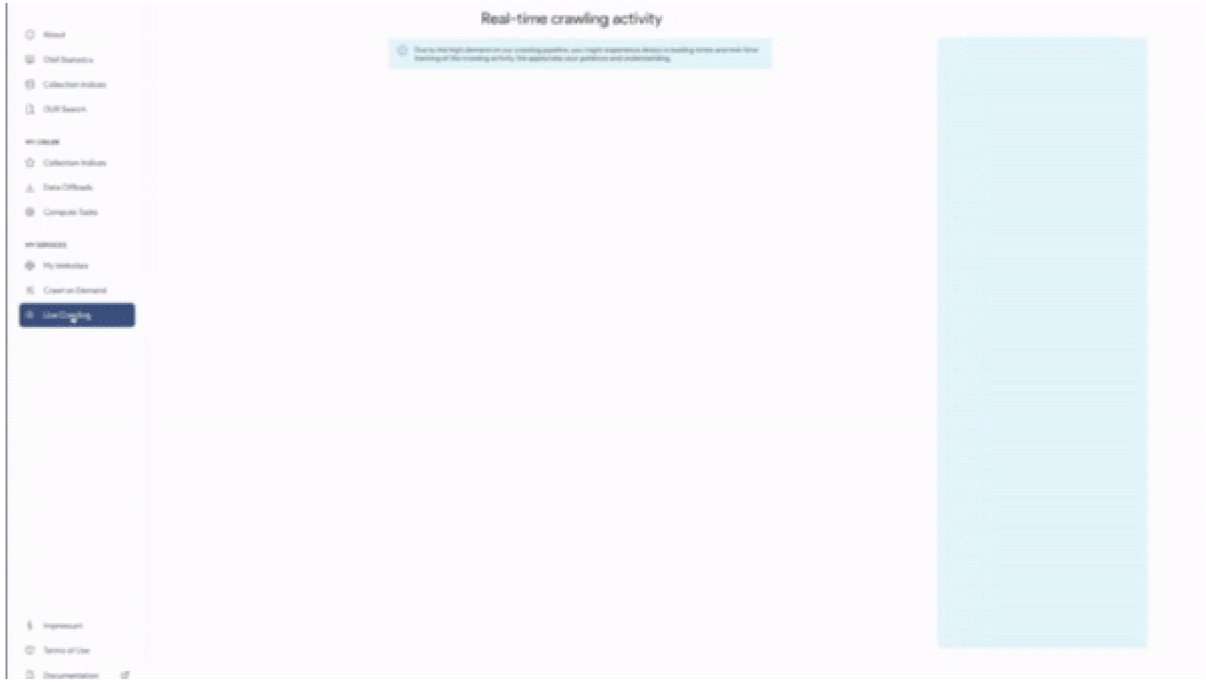
Index & Federation

- **~1.1 PB** raw data · **~50 TB** index data · **810+ public datasets**
- **10–20 TB** new index data per month
- Approx. **factor 20** smaller than commercial indices (factor 10–50× for multimedia)

Federated Architecture

- Distributed over **IT4I, CSC, LRZ, CERN** — onboarding further AI Factories
- Cross-DC workflow execution via LEXIS platform
- Federated storage: iRODS + S3; client-side SQL access via owilix / DuckDB
- Dense embeddings available on select slices

Step 1: Crawling the Web



- 100 Million URLs/day across 25 machines in 3-4 data centers
- Central coordinator/queue at CSC/CERN, 2 billion URLs in queue
- Focus on European domains and European languages
- Respects robots.txt, Gen AI opt-out directives, no-index
- Filters potentially harmful and low-quality content

All systems are online

All components are functioning properly. Data repositories and services are accessible.

Last Updated: 18/06/25, 06:15

SYSTEMS	DESCRIPTION	COMPONENTS	STATUS
IT4I	connected	Repository, Project, Public	ONLINE
LRZ	connected	Repository, Project, Public	ONLINE

Crawled Data
1037.57 TB

+75.75TB last month

Jun 1, 2025 last updated

WARC Datasets
174

+23 last month

Jun 1, 2025 last updated

Public Datasets
655

+165 last month

N/A last updated

Open Web Index
28.83 TB

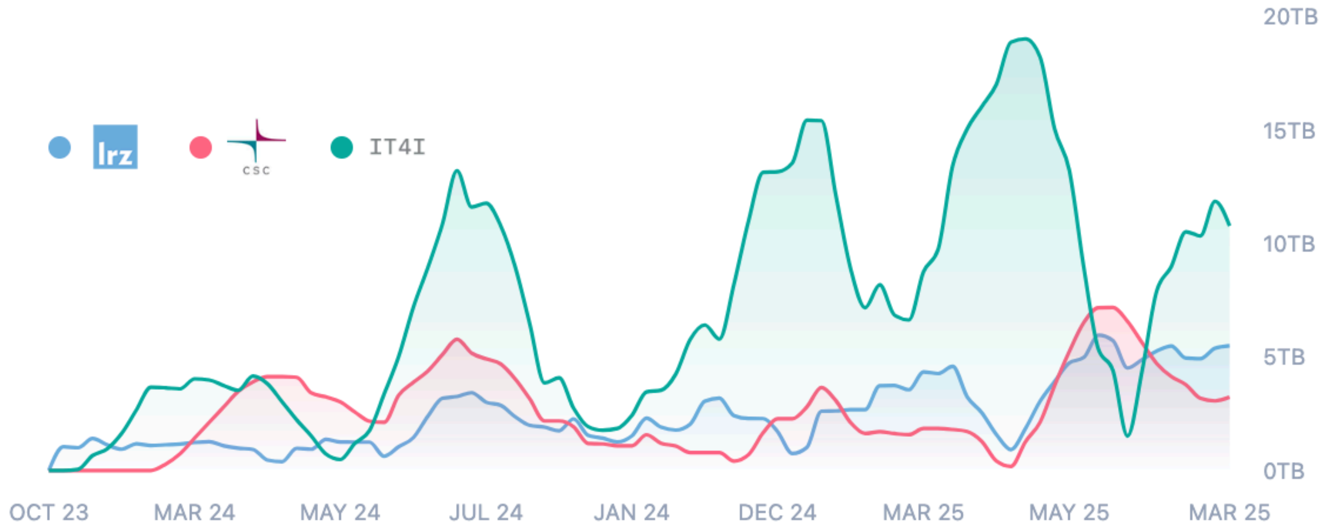
+2.1TB last month

N/A last updated

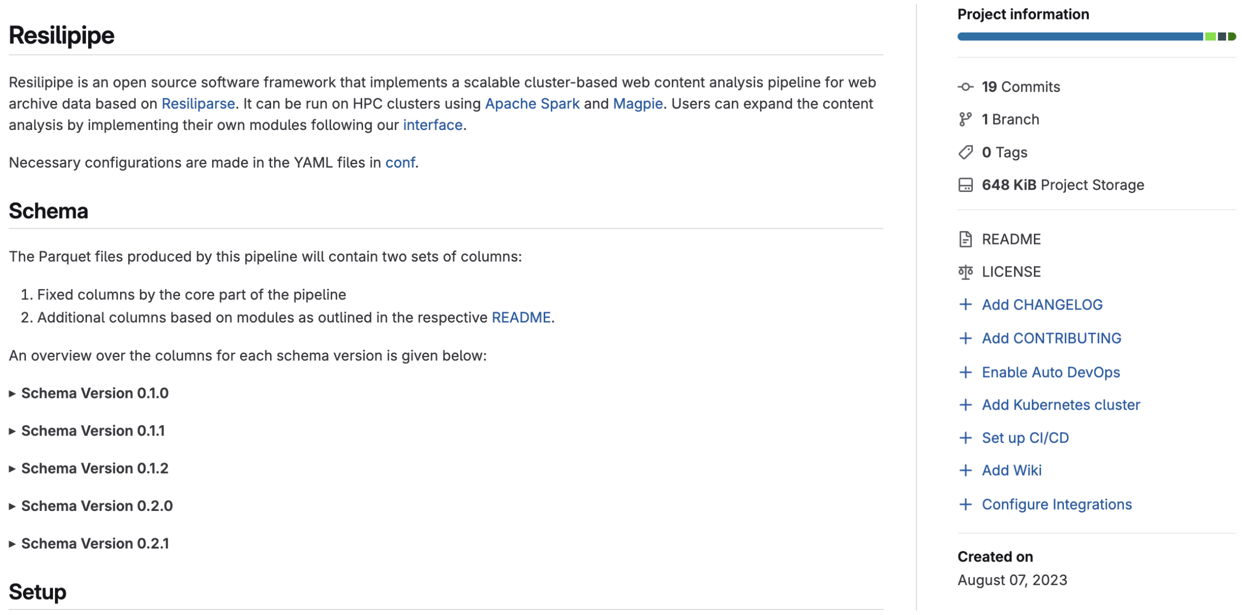
Language Distribution

- English 38%
- German 8%
- French 6%
- Spanish 6%
- Chinese 6%
- Others 36%

Crawl Volume by Datacenter



Step 2: Preprocessing



Resilipipe

Resilipipe is an open source software framework that implements a scalable cluster-based web content analysis pipeline for web archive data based on Resiliparse. It can be run on HPC clusters using Apache Spark and Magpie. Users can expand the content analysis by implementing their own modules following our interface.

Necessary configurations are made in the YAML files in conf.

Schema

The Parquet files produced by this pipeline will contain two sets of columns:

1. Fixed columns by the core part of the pipeline
2. Additional columns based on modules as outlined in the respective README.

An overview over the columns for each schema version is given below:

▸ **Schema Version 0.1.0**

▸ **Schema Version 0.1.1**

▸ **Schema Version 0.1.2**

▸ **Schema Version 0.2.0**

▸ **Schema Version 0.2.1**

Setup

Project information

19 Commits
1 Branch
0 Tags
648 KiB Project Storage

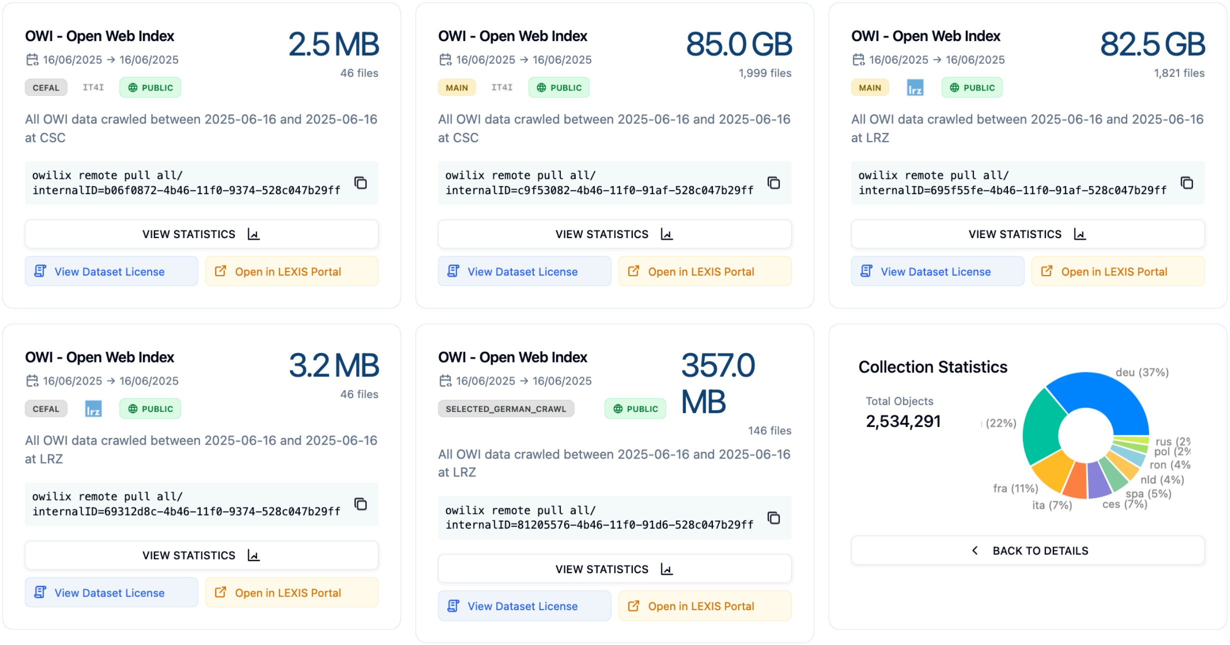
README
LICENSE
+ Add CHANGELOG
+ Add CONTRIBUTING
+ Enable Auto DevOps
+ Add Kubernetes cluster
+ Set up CI/CD
+ Add Wiki
+ Configure Integrations

Created on
August 07, 2023

Resilipipe — open-source, modular, scalable pipeline

- HTML to cleaned text at scale (reduction ratio 1:5)
- **Metadata extraction**: JSON-LD, Microdata, OpenGraph
- **Semantic enrichment**: language, topics, geo-locations, NER
- **Structured data extraction** from 60% of web pages: FAQ, opening hours, products, job postings
- Open to community contributions via plugin architecture

Step 3: Building and Distributing the Index

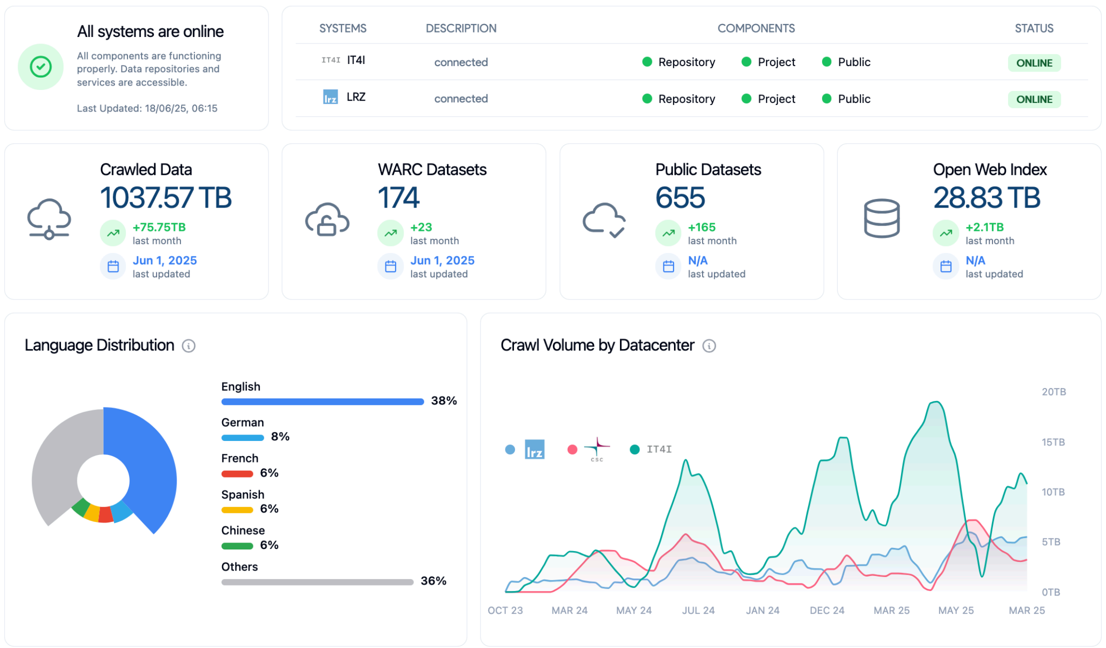


- One dataset per day per data center
- **Sparse full-text index**: CIFF format, SQL-based querying
- **Feature index**: link structure, linguistic style, genre, metadata
- **Dense index**: embeddings (in development), compression techniques for ANN search
- Pre-computed collection indices: curlie, legal, main, metadata/microdata
- Tooling for download and dataset management via Owilix

```
OWI Dataset

- Year=2023
  - month=12

    - day=24

      -language=eng

        index.ciff.gz
        metadata_#.paruet

      -language=....
```

Accessing the Open Web Index

Dashboard — Statistics & Datasets in Detail



OWI Statistics Tab

- Repository health per data center (IT4I · LRZ)
- Key indicators: pipeline runs, HPC core-hours consumed, avg. duration, active pipelines
- Crawled data (1.3 PB), WARC datasets, public datasets, index size
- Language distribution chart + crawl volume timeline per DC
- Dataset collection distribution by resource type (main / GPU / curlie / embeddings)

OWI Datasets Tab

- Filter: datacenter · format (WARC / Parquet / CIFF / Embeddings) · language · date range
- Infinite-scroll dataset cards: title, date, DC, public/private badge, file listing
- Parquet file preview in-browser; download via LEXIS Portal or `owilix remote pull`

Dashboard — Corpora, Models & Search

OWI Corpora

Curated, administrator-prepared collections for distribution:

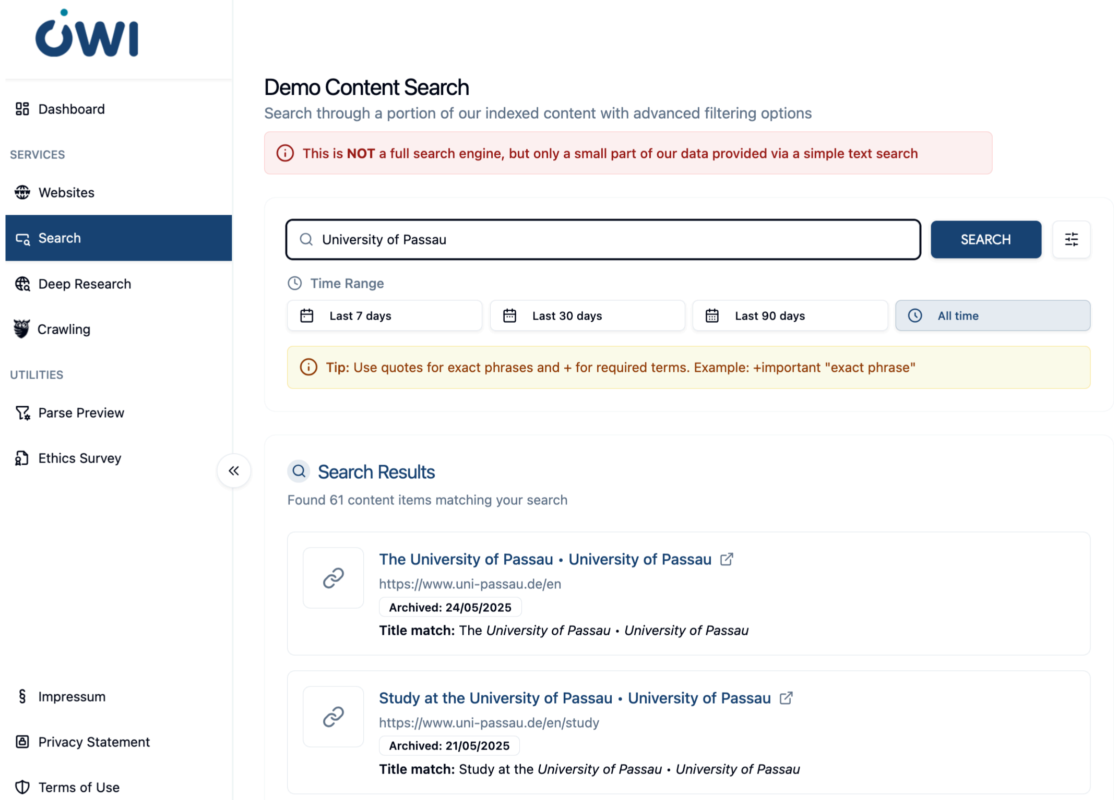
- Licensing metadata per corpus (creator, terms, full licence text)
- LEXIS integration for storage, access control and metadata
- Includes domain-specific and project-specific collections

OWI Models

Registry of language models for web search:

- Browser-runnable models via WebAssembly (Wllama library)
- Cloud-based models (OpenAI, DeepSeek) for heavy tasks
- Model cards with training data, benchmarks, and GGUF file listing

OUR Search (Demo)

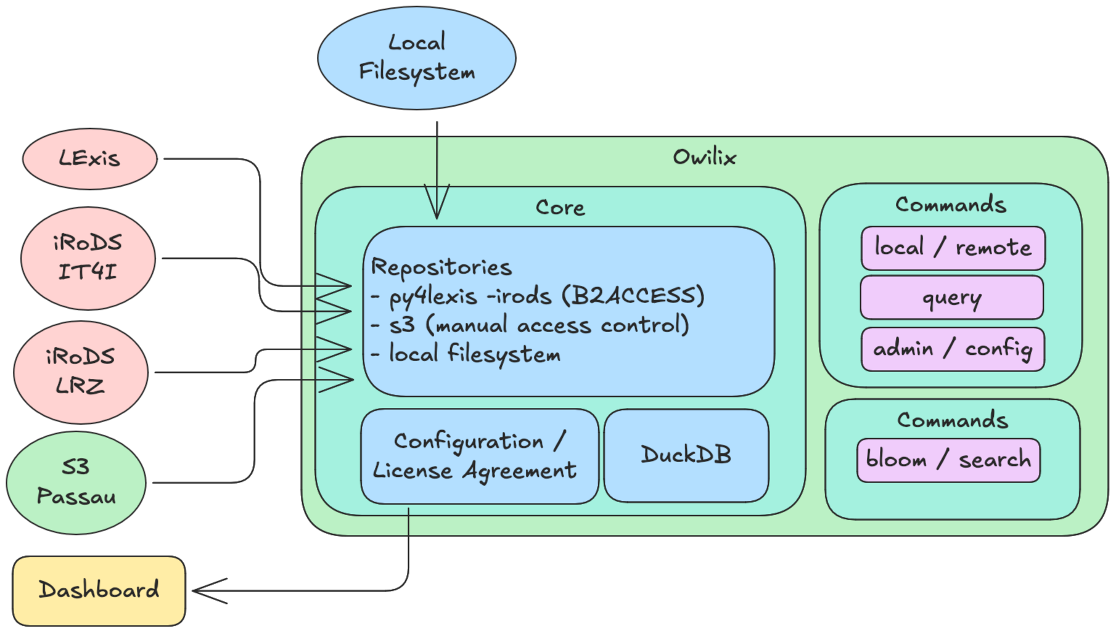


Simple + advanced field-specific search over a subset of the index; time-range filtering; result cards with thumbnail, title, URL

Deep Research (Experimental)

AI-powered multi-step research: configurable depth + breadth, local or cloud LLM, interactive clarification mode

owilix — The Open Web Index Client



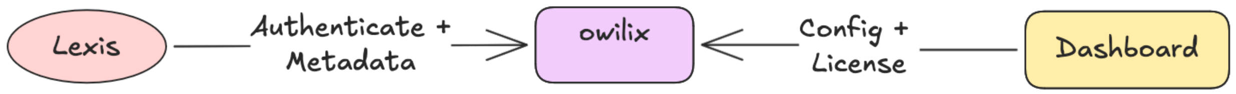
Four main workflows:

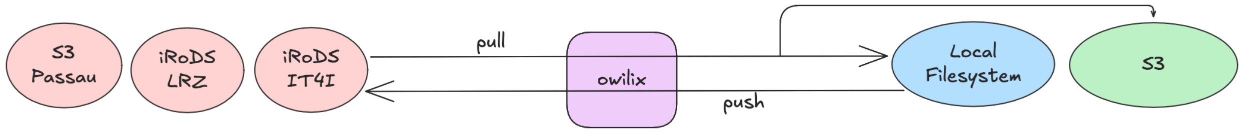
1. **Authentication** — B2ACCESS via LEXIS + Dashboard config (OpenID Connect / Keycloak)
2. **Sync Datasets** — pull/push between federated iRODS/S3 storage and local filesystem
3. **Query / Slice** — client-side SQL queries with JSON, HTTP, OpenSearch output
4. **WARC Access** — fetch original WARC files via SQL querying

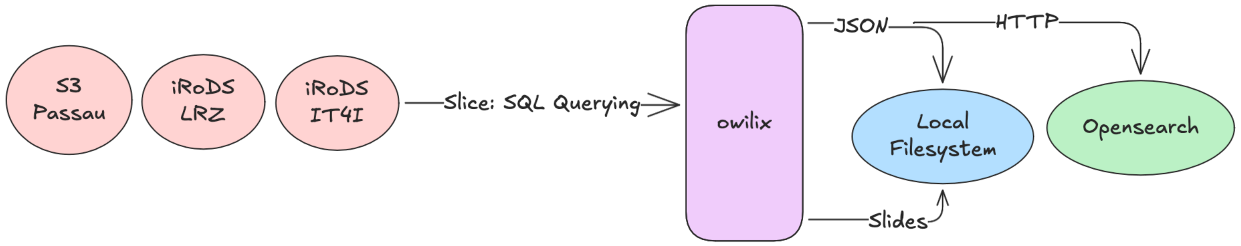
Slice the index by:

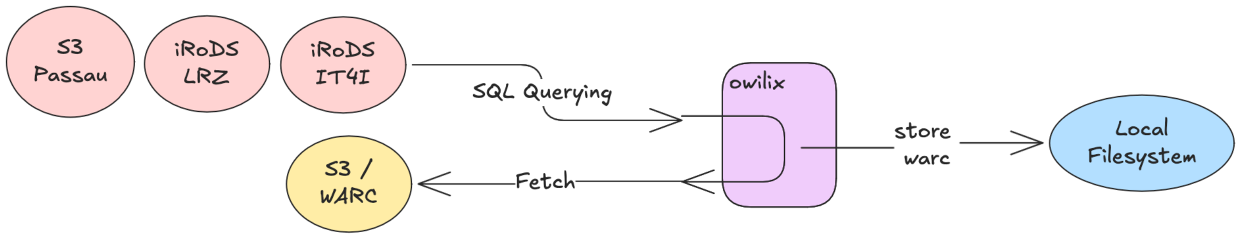
- **Domain** (.de, .at, .fr, …)
- **Topic** (Curlie.org hierarchy labels)
- **Language** (filename-based selection)
- **Sitelist** (CSV of URLs / domains / TLDs)
- **Structured data** / microdata type
- **Outlinks** for link graph analysis

owilix Workflows









Remote Index — Query Without Downloading

What is the remote index?

A DuckDB / Apache Spark-queryable representation of the OWI — Parquet partitions aligned with CIFF shards, accessible from remote clients without downloading the full index.

Example queries

```sql
-- Count documents per language
SELECT language, COUNT(*) as docs
FROM read_parquet('remote://lrz:latest/main/**')
GROUP BY language ORDER BY docs DESC;

-- Filter by domain
SELECT url, title FROM ... WHERE domain = 'wikipedia.org';
```

Use cases

- Analytics and statistics on the live index
- Lightweight RAG pipeline ingestion
- Ad-hoc exploration before committing to a full shard download

OUR²S — Online Search API

OUR²S (built on Vespa) extends the remote index with a full REST search API:

- Structured document model: full page · snippets · embeddings
- Keyword, semantic and hybrid ranking modes
- Multi-index federation across data centers
- No local index needed — query directly over network

> → OUR²S details in the dedicated OUR²S column

Access summary

Mode	Tool	Authentication
Browse	Dashboard	Anonymous / B2ACCESS

Mode	Tool	Authentication
Download	owilix	B2ACCESS (LEXIS)
SQL queries	DuckDB remote	B2ACCESS
REST search	OUR²S API	API key
Conversational RAG	MosaicRAG	Open / B2ACCESS

MosaicRAG — Building RAG Systems on OWI

From OWI Datasets to Conversational Search

MosaicRAG consumes OWI daily datasets directly — CIFF inverted index files, Parquet metadata records, and dense embeddings (published since Dec 2025) — and exposes them through a modular conversational interface.

Pipeline for building your own MosaicRAG instance:

1. Download OWI slice with `owilix` (topic, domain, or language filter)
2. Ingest CIFF → inverted index; Parquet → DuckDB; embeddings → ChromaDB
3. Configure retrieval pipeline: sparse, dense, or hybrid
4. Choose reranker (TF-IDF · Embedding cross-encoder · Tournament LLM · Group LLM)
5. Optionally add summarization and relevance marking

> Full documentation: **openwebindex.eu/book** — "Building vertical search engines with MOSAIC"

Supported LLMs

Task	Models
Reranking	gemma2, qwen2.5, llama3.1
Summarization	+ DeepSeekv3
Relevance marking	gemma2, qwen2.5, llama3.1

Live Demo

Try conversational search on the Open Web Index at

mosaicrag.ows.eu

Using the Open Web Index — Applications

Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → mosaic.ows.eu
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index → args.me/owi

Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems, and domain-specific applications

MosaicRAG — Conversational Search on the OWI

Modular RAG framework extending MOSAIC with LLM-based capabilities and dense retrieval (WEBIS / TU Graz)

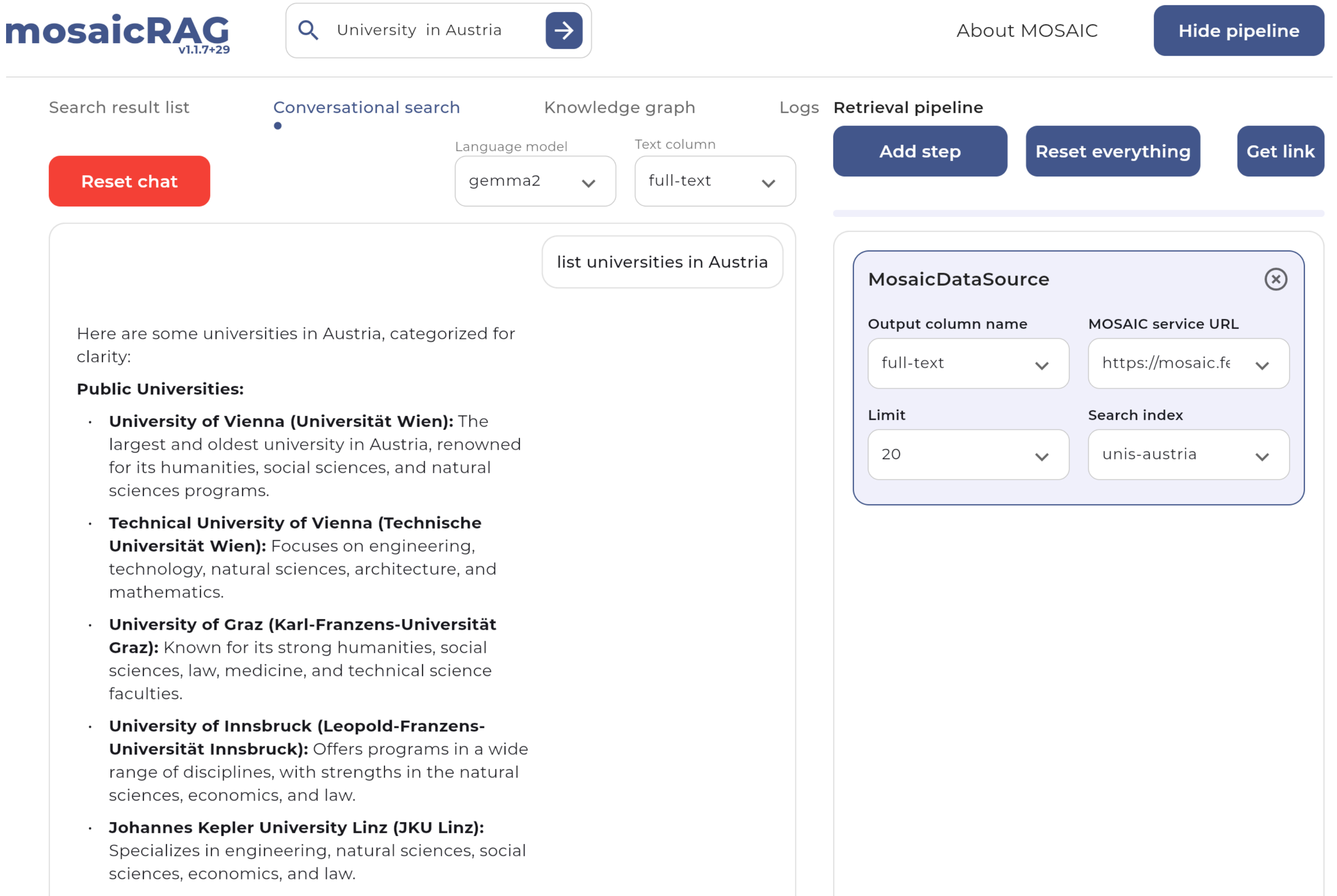
- **Two retrieval paths:** inverted index (CIFF) + dense retrieval
- **4 reranking strategies:** TF-IDF, Embedding cross-encoder, Tournament-style (LLM pairwise), Group-style (LLM window)
- **LLM backends:** self-hosted on-premise models for summarization
- **Conversational search:** follow-up questions about retrieval results; dual-window UI (search results + chat)
- **Workflow:** define retrieval pipeline → issue query → converse about results

Key Features

- Relevance marking — LLM highlights relevant passages within documents
- Summarization — user-provided instructions applied per document
- Domain-specific indices: science, health, arts (8M+ docs)
- Fully open-source modular pipeline

Data Pipeline

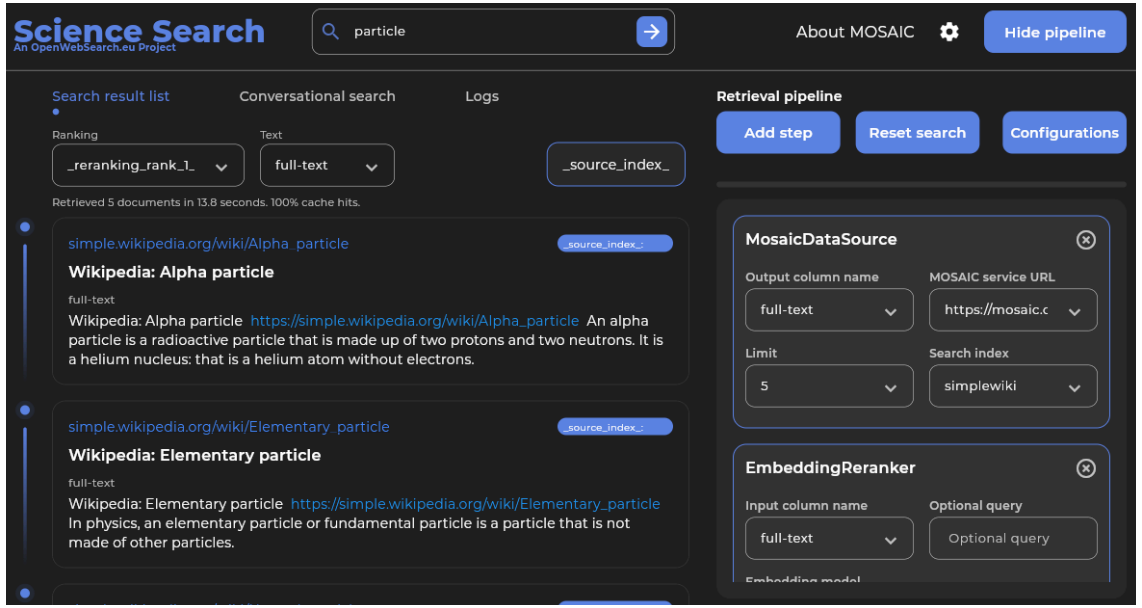
OWI daily datasets → CIFF (inverted index) + Parquet (metadata) + dense embeddings (since Dec 2025) → MosaicRAG



Demo: mosaicrag.ows.eu · mosaic.ows.eu

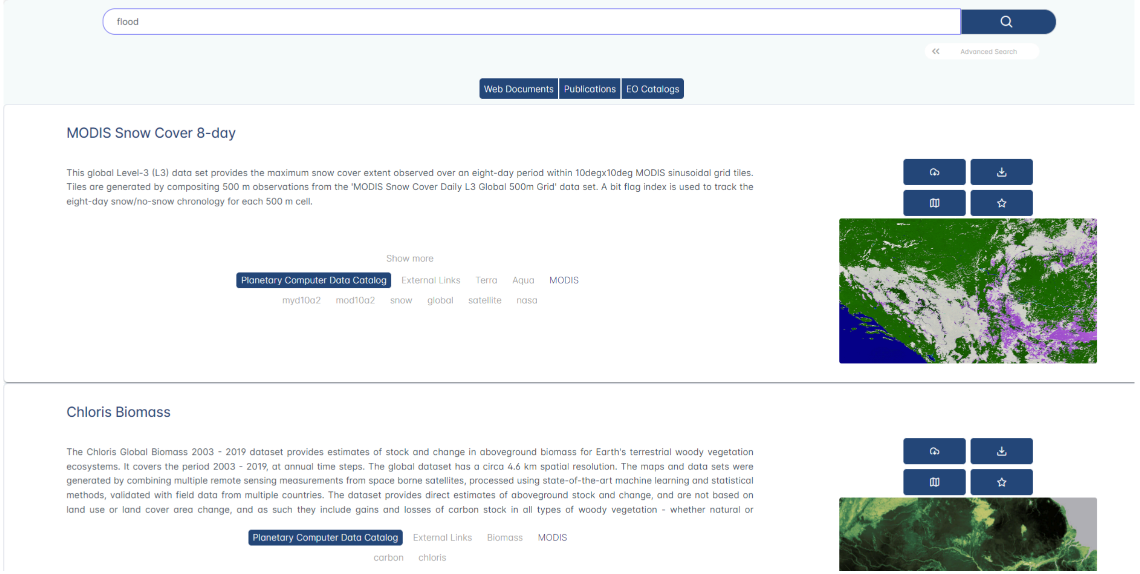
Mosaic Search Applications · Earth Observation Search

Mosaic Vertical Search (TU Graz)
Modular framework for building vertical search engines on OWI slices — domain-specific indices for science, health, arts (8M+ docs)



Demo: mosaic.ows.eu · mosaic.ows.eu/demos

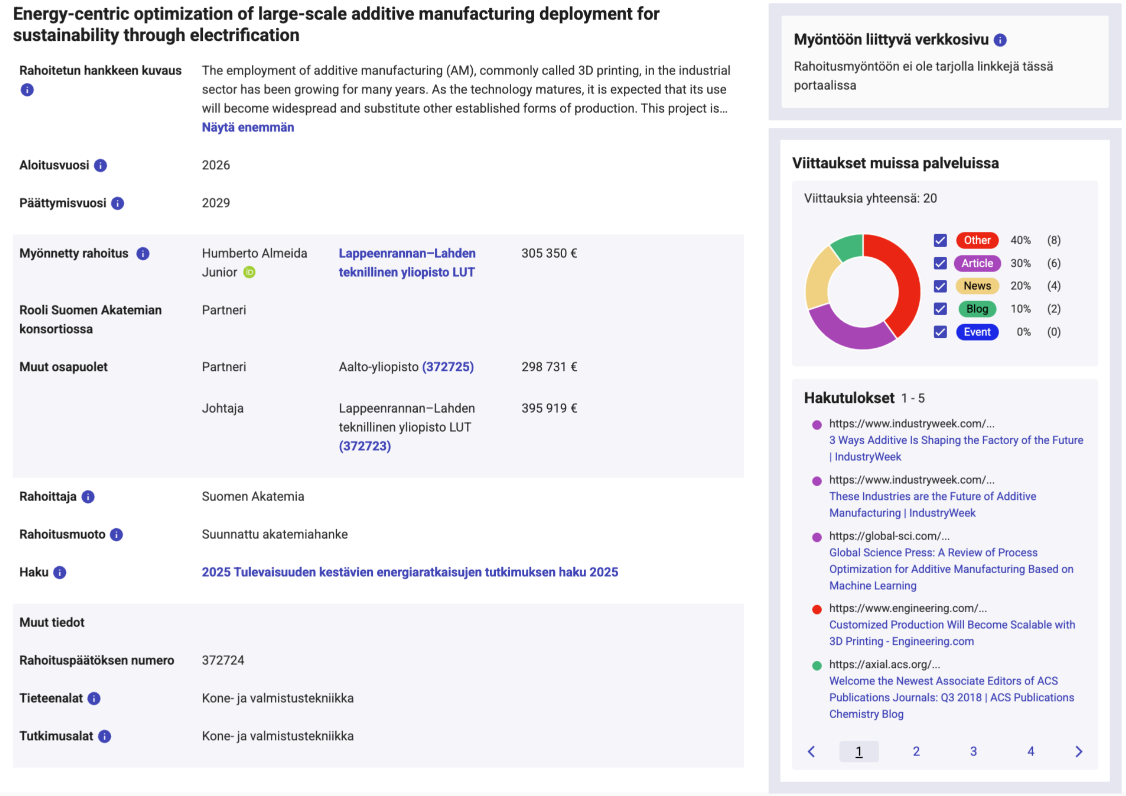
Earth Observation Search (DLR / Radboud)
RAG-based QA over scientific papers, datasets & web content; geo-contextualized multi-genre search on interactive maps; KG with 2M+ nodes (OpenAlex, PANGAEA, DLR Geoservice)



Demo: eo-rag-demo.2.rahtiapp.fi · ows-demo.2.rahtiapp.fi

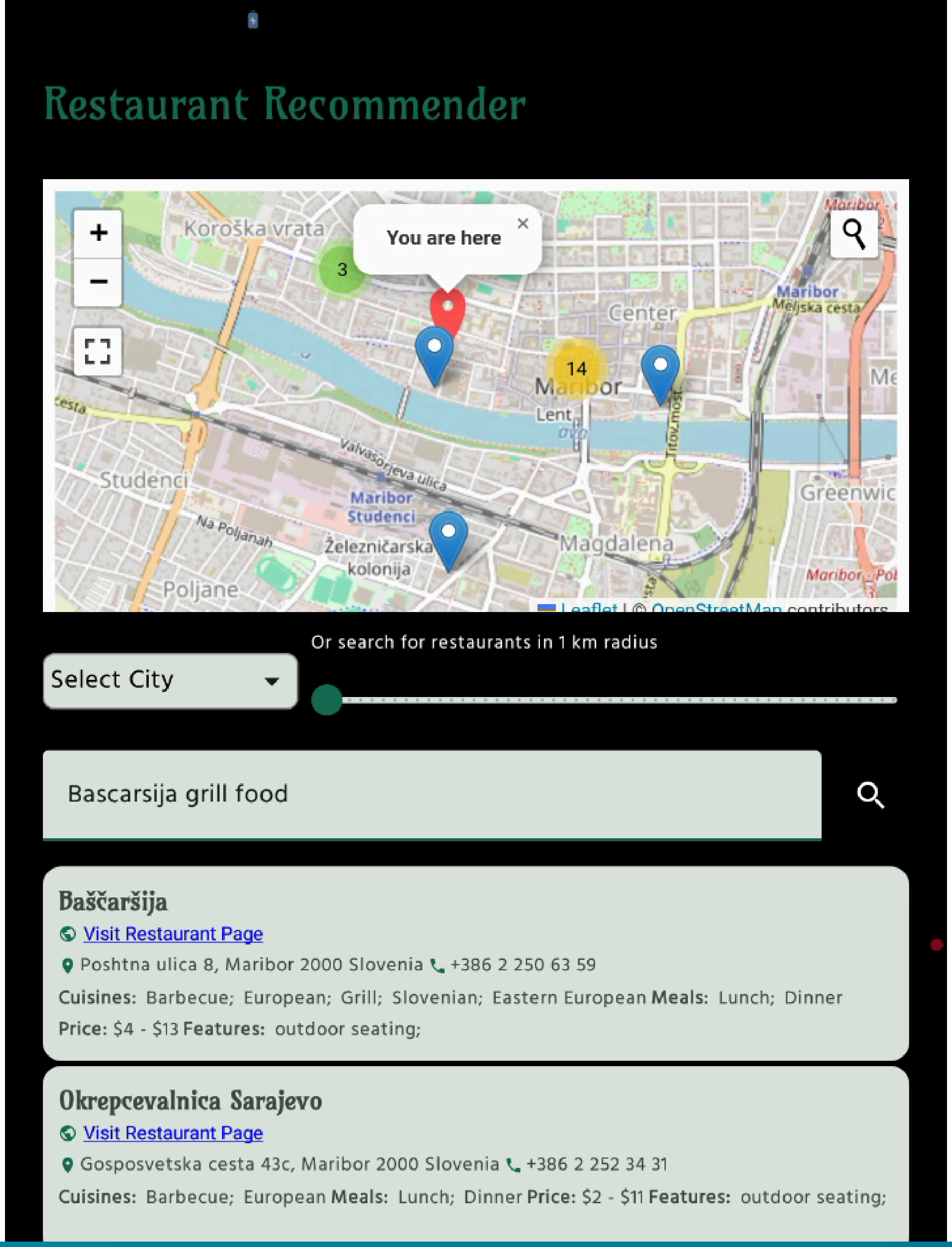
Multilingual Scientific Index · Restaurant Recommender

Multilingual Scientific Index (CSC)
Enriches Finland's Research.fi portal with OWI web data — Finnish & English science content classified by type (publication, blog, event); embedded directly in national research portal



Demo: research.fi

Restaurant Recommender (A1 Slovenia)
Privacy-preserving mobile app; on-device personalization via query imputation; LLM-extracted restaurant features from OWI; 5-month pilot with 214 users — overall satisfaction 4.2 / 5



Restaurant Recommender
You are here
3
14
Koroška vrata
Center
Maribor
Lent
Studenci
Valvasorjeva ulica
Maribor Studenci
Na Poljanah
Železničarska kolonija
Magdalena
Poljane
Greenwic
Leaflet | © OpenStreetMap contributors
Or search for restaurants in 1 km radius
Select City
Bascarsija grill food
Baščaršija
Visit Restaurant Page
Poshtna ulica 8, Maribor 2000 Slovenia +386 2 250 63 59
Cuisines: Barbecue; European; Grill; Slovenian; Eastern European Meals: Lunch; Dinner
Price: $4 - $13 Features: outdoor seating;
Okrepcevalnica Sarajevo
Visit Restaurant Page
Gosposvetska cesta 43c, Maribor 2000 Slovenia +386 2 252 34 31
Cuisines: Barbecue; European Meals: Lunch; Dinner Price: $2 - $11 Features: outdoor seating;

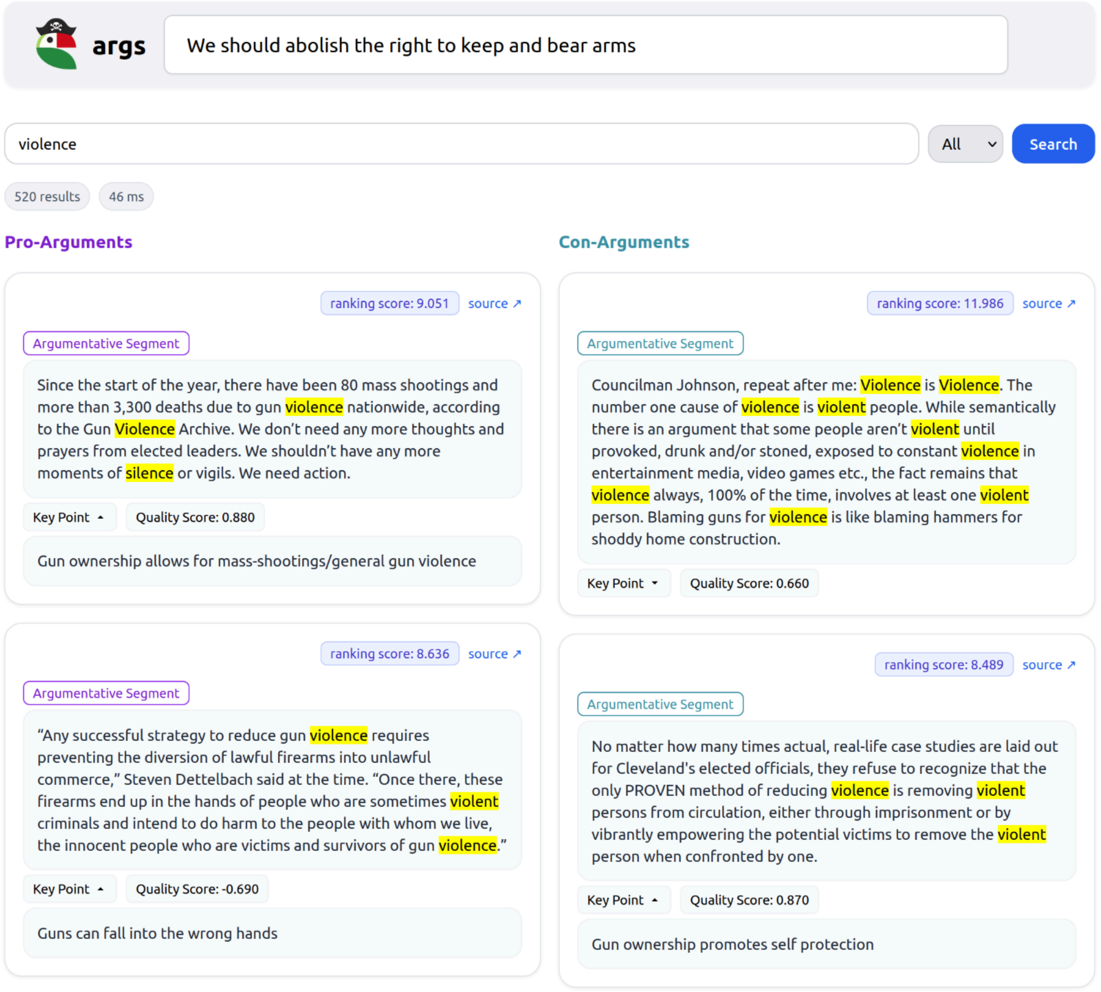
Argumentation Search · CERN Search Applications

Argumentation Search (WEBIS)

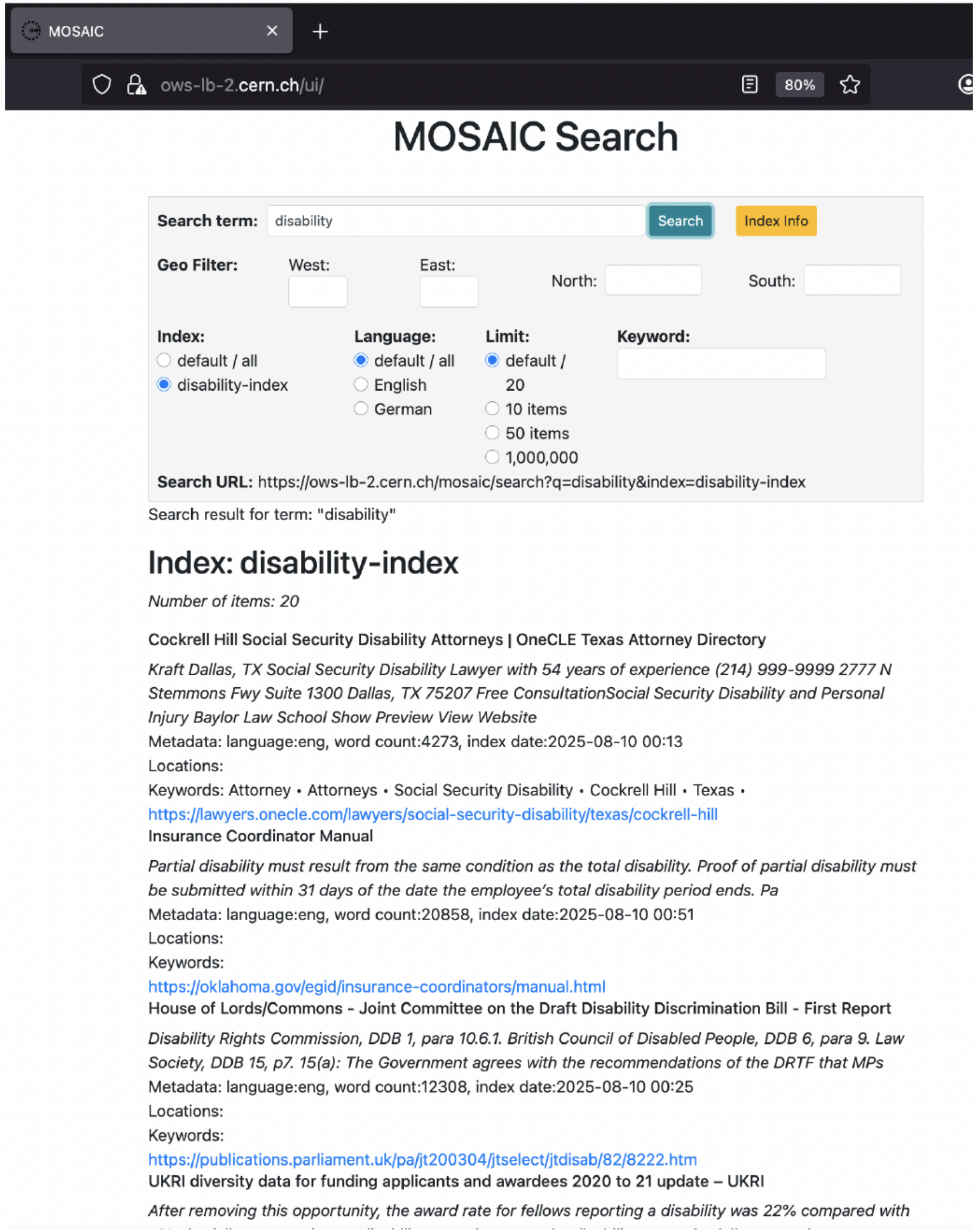
args.me extended with OWI: pro & contra arguments on 31 controversial topics; argument quality scoring, key point matching, temporal trend analysis

CERN Search Applications

- **Institutional**: 25K+ pages crawled from CERN internal web; full OWS pipeline → MOSAIC index for AccGPT knowledge base
- **Nooon**: disability knowledge search (2.59M docs) for HR & Diversity & Inclusion



Demo: args.me/owi · args.me

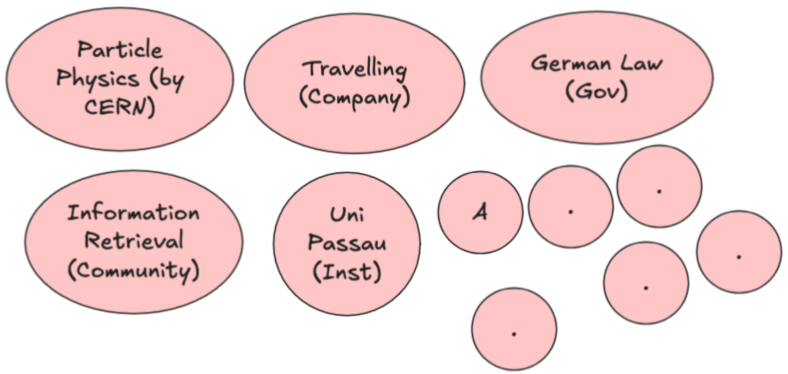


Deployed on open-science-search.cern.ch

Third-Party Projects (FSTPs)

- **AKSE** — argumentation knowledge graph for structured deliberation
- **VERITAS / DEXAI** — verification-oriented RAG for sensitive geopolitical contexts (Ukraine)
- **OMMS** — enriching OpenStreetMap POI with OWI structured data
- **DTCommerce** — product extraction & enrichment pipeline for SME e-commerce
- **CIFFIL** — cross-collection index interoperability for municipal search (Dutch pilots)
- **TILDE** — trustworthy health search with bias detection and entity-level retrieval

Towards Federated RAG Systems



- RAG / Agentic systems require focused, high-quality collections
- Web information redundancy increases due to LLMs — no one-size-fits-all retrieval
- OpenWebSearch.eu supports low-cost creation of topic-specific RAG systems

Key ideas:

- Retrieve from well-designed RAG systems instead of generic search engines
- Topic-specific, deduplicated and curated content
- Domain-optimized ranking, aggregation and summarization
- Lightweight browser-based federation with personal data integration

> **OwlerLite** (WWW 2026) — privacy-respecting browser extension for scope- and freshness-aware retrieval: user-defined **scopes** (trusted sources), semantic

freshness detection via SimHash + vector similarity, Scope Fidelity improved from 0.64 → **0.83** (negligible NDCG loss)

Using the Open Web Index — Data & Analytics

Published Datasets

Dataset	Scale	Use case
WebFAQ	96M Q&A, 75 langs	QA training, cross-lingual IR
WebFAQ 2.0	198M pairs, 108 langs	Dense retrieval, hard negatives
Imprints	5.25M hosts, geocoded	Enterprise analytics, Eurostat
German Commons	154B tokens	German LLM pre-training
OWS-Curlie-2025	20.7M docs + qrels	IR evaluation benchmark

All datasets available at openwebindex.eu/corpora

Analytics Potential

- **Official Statistics**: enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring**: supply chain analysis, news monitoring (JRC)
- **AI Training Data**: domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks**: web-scale IR test collections with relevance assessments
- **Trend Analysis**: temporal web monitoring at Petabyte scale

> The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

WebFAQ — Multilingual FAQ from the Web

- **96 million** Q&A pairs across **75+ languages**
- Extracted from structured FAQ markup (JSON-LD, Microdata) in the Open Web Index
- Fields: question, answer, URL, origin, topic, question type
- Parquet format, hosted on Hugging Face

Use cases: multilingual QA training, information retrieval, cross-lingual transfer learning

huggingface.co/datasets/PaDaS-Lab/webfaq

Top languages	Pairs
English	~30M
German	~8M
French	~6M
Spanish	~5M
...	75+ langs

WebFAQ 2.0 — Hard Negatives for Dense Retrieval

- **198 million** FAQ-based Q&A pairs across **108 languages**
- **14.3 million** bilingual aligned QA pairs — largest FAQ-based resource
- **1.25 million** queries with mined hard negatives (20 languages)
- 200 cross-encoder scored negatives per query

Key contributions:

1. Novel web crawling for diverse content extraction
2. Hard negatives for dense retriever training
3. Contrastive Learning & Knowledge Distillation
4. Open datasets and training scripts

Dinzinger, Caspari, Salman, Topi, Mitrović & Granitzer. "WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval." arXiv:2602.17327, 2025.

Imprints Dataset — Geospatially Grounded Web Curation

- **5.25 million** hosts, **47.8 GB**
- Each website annotated along two dimensions:
  - **Topical classification** via LLM (Foursquare category hierarchy)
  - **Geographic grounding** from legal imprints via geocoding
- **3.08 million** websites geocoded from extracted addresses
- Page types: main page, imprint, contact, about us, privacy, terms

Country	Geocoded	Share
Germany	2,636,913	85.5%
Austria	223,805	7.3%
Switzerland	160,683	5.2%

Top-Level Topic	Share
Business & Professional	41.6%
Community & Government	16.2%
Arts & Entertainment	12.2%

Dinzinger, Rittlinger, Voigt, Gadziomski, Granitzer, Geiß & Taubenböck. Uni Passau & DLR. *openwebindex.eu/corpora*

German Commons — 154B Tokens for German LLMs

- **154.56 billion tokens**, **35.78 million documents**
- Aggregated from **41 diverse sources** under open licenses
- Quality-filtered, deduplicated, OCR scored

Domain	Tokens
News Commons	72.67B
Cultural Commons	54.49B
Web Commons	19.89B
Political Commons	3.57B
Legal Commons	2.99B
Scientific Commons	0.84B
Economics Commons	0.11B

Gienapp, Schröder, Schweter et al. "The German Commons — 154 Billion Tokens of Openly Licensed Text for German Language Models." arXiv:2510.13996, 2025.

OWS-Curlie-2025 — Web Search Test Collection

- **20.7 million** documents from 6 months of OWI crawl data (March–August 2025)
- English documents from the Curlie subcollection (websites labeled in the Curlie directory)
- Three evaluation splits with student-provided topics and relevance assessments
- Jina V3 embeddings, CIFF indexes for sparse retrieval
- Accessible via `owilix` CLI and `ir-datasets-owi` Python package

Split	Documents	Embeddings	Qrels
`all`	20,746,181	--	--
`radboud-val`	63,639	yes	yes
`radboud`	90,852	yes	--
`jena-kassel`	77,382	yes	--

Ethical, Legal & Societal Aspects (ELSA)

ELSA Framework — Values, Risks & Mitigation

10 Ethical Values

1. Transparency
2. Safety & Security
3. Openness
4. Responsibility
5. Privacy
6. Justice
7. Trustworthiness
8. Accountability
9. Sustainability
10. Sovereignty / Autonomy

5 Ethical Risks

1. Control of indexed websites & take-down requests — censorship vs. coverage
2. Flawed content in the OWI — false, misleading, offensive material
3. Inaccurate metadata — errors in ingestion or preprocessing
4. Misuse by third parties — privacy violations, unethical applications
5. Unsuitable governance — undemocratic control, security breaches

ELSA Deliverables

Three iterations (D6.1 → D6.2 → D6.3), final version subsumes all. Developed with:

- 3 multi-day ethics workshops (CAIS / Stiftung Mercator)
- 3 legal FSTP projects (LISA, LOREN, ALMASTIC)
- OSF Working Groups on Ethics & Law

6 Stakeholder Groups

End users · Index users · Developers · Researchers & institutions · Content providers · Governance

Legal Challenges & Regulatory Landscape

Regulatory Framework

- **8+ EU regulations** apply: GDPR, Copyright Directive, DSM Directive, DSA, DMA, Data Act, Data Governance Act, AI Act
- **Copyright is the biggest challenge** — impacts crawling, redistribution, and LLM training in Europe
- No guiding court decisions yet — first time an open web index operates in the EU
- DSA classification unclear — OWI does not fit neatly as hosting service, platform, or search engine
- **AI Act**: OWI classified as **"limited risk"** — research exemption applies during project; post-project needs reassessment

Legal FSTP Projects (published on Zenodo)

- **LISA** — Legal Framework for OWI (Matthias Wendland)
- **LOREN** — Legal Open European Web Index (Paul Johannes, Maxi Nebel)
- **ALMASTIC** — Assessing Legal Risks (Kai Erenli, Andreas Gruber)

Key Legal Risks

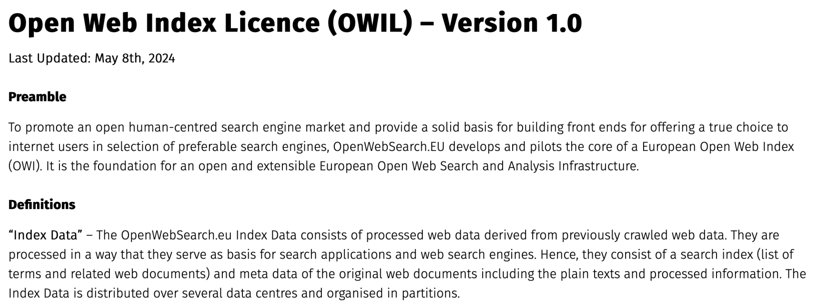
- Personal data in crawled content (GDPR Art. 6(1)(f) legitimate interest)
- Copyright infringement — images, paywalled articles, plain text
- Liability as digital intermediary
- Illegal content obligations
- LLM training restrictions (no_genai flag)

10 Technical & Organisational Mitigation Measures

#	Measure	Key Mechanism
M1	Website Management	robots.txt opt-out, blacklisting, quality prioritisation
M2	Take-down Management	Dashboard submission, logged removal, blacklist propagation
M3	Content Analysis	Quality scoring, trigger warnings, ad detection, plugin architecture
M4	Access Control	B2ACCESS registration, immediate blocking of misuse
M5	Transparency Guideline	27-aspect guideline for third-party search apps
M6	Data Protection & Privacy	Anonymised queries, minimal user data, reusable model
M7	Safety & Security	Regular security meetings, incident reporting
M8	OWI Licence	Research (default) · Commercial (OWICL) · Data Access (raw WARC)
M9	Legal & Ethical Self-Assessment	Questionnaire at download time, ethics score, signed PDF certificate
M10	Ethical & Legal Advice	Ethics workshops, legal FSTPs, OSF working groups

Three Licence Types

- **Research** — default; included in owilix
- **Commercial (OWICL)** — requires personal contact with coordinator
- **Data Access** — raw WARC; research only, personal request



Open Web Index Licence (OWIL) – Version 1.0

Last Updated: May 8th, 2024

Preamble

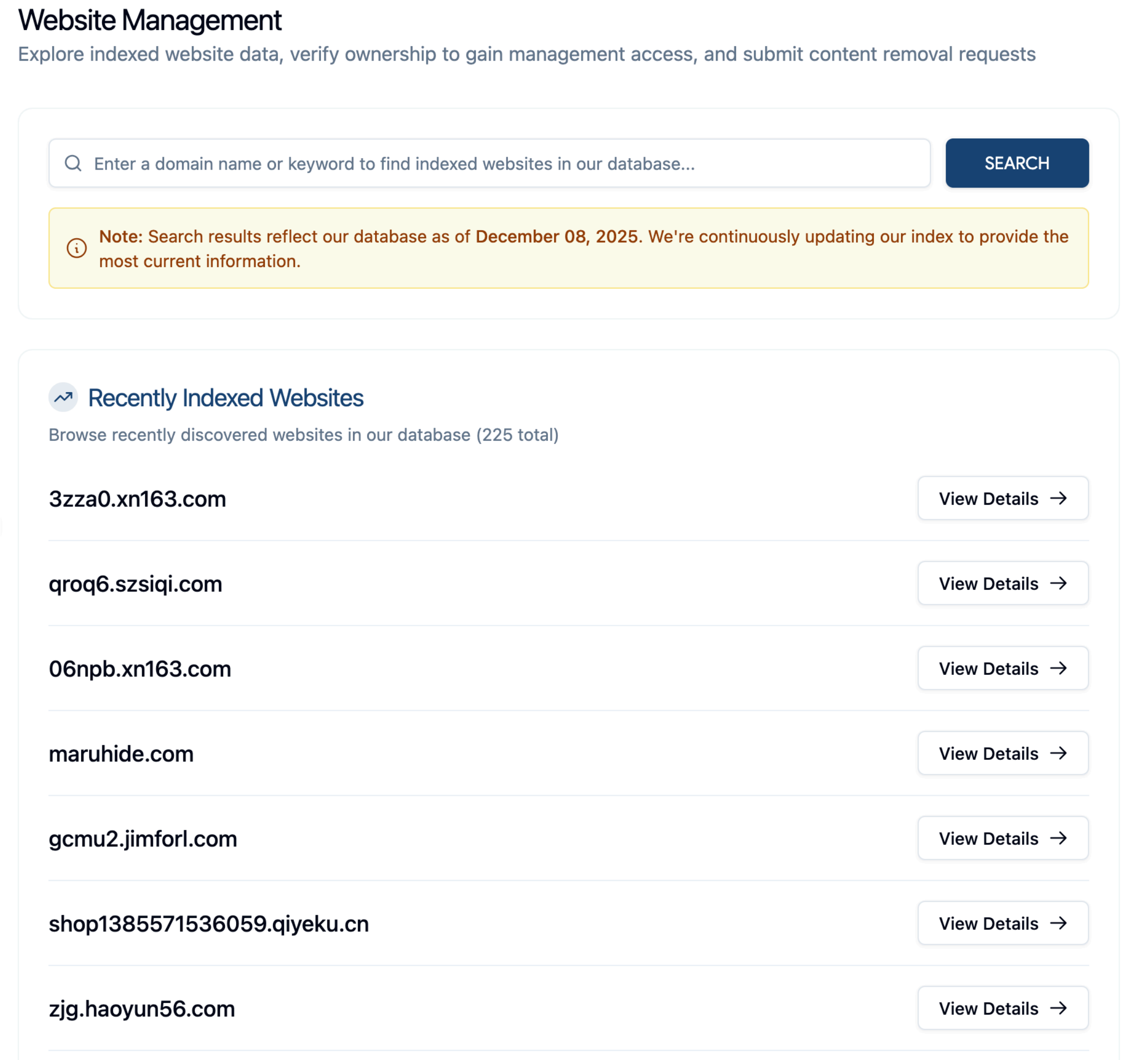
To promote an open human-centred search engine market and provide a solid basis for building front ends for offering a true choice to internet users in selection of preferable search engines, OpenWebSearch.EU develops and pilots the core of a European Open Web Index (OWI). It is the foundation for an open and extensible European Open Web Search and Analysis Infrastructure.

Definitions

"Index Data" – The OpenWebSearch.eu Index Data consists of processed web data derived from previously crawled web data. They are processed in a way that they serve as basis for search applications and web search engines. Hence, they consist of a search index (list of terms and related web documents) and meta data of the original web documents including the plain texts and processed information. The Index Data is distributed over several data centres and organised in partitions.

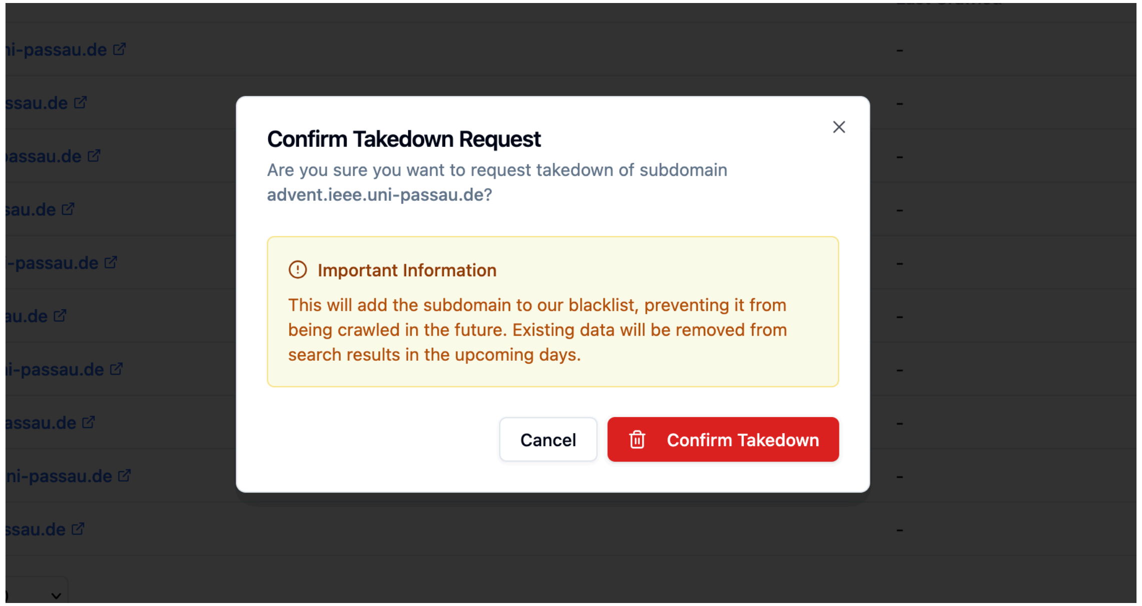
Open Webmaster Console — Transparency Tools

Website Management



- Search any domain in the OWI — crawl stats, content types, language distribution,

Content Removal & Rights



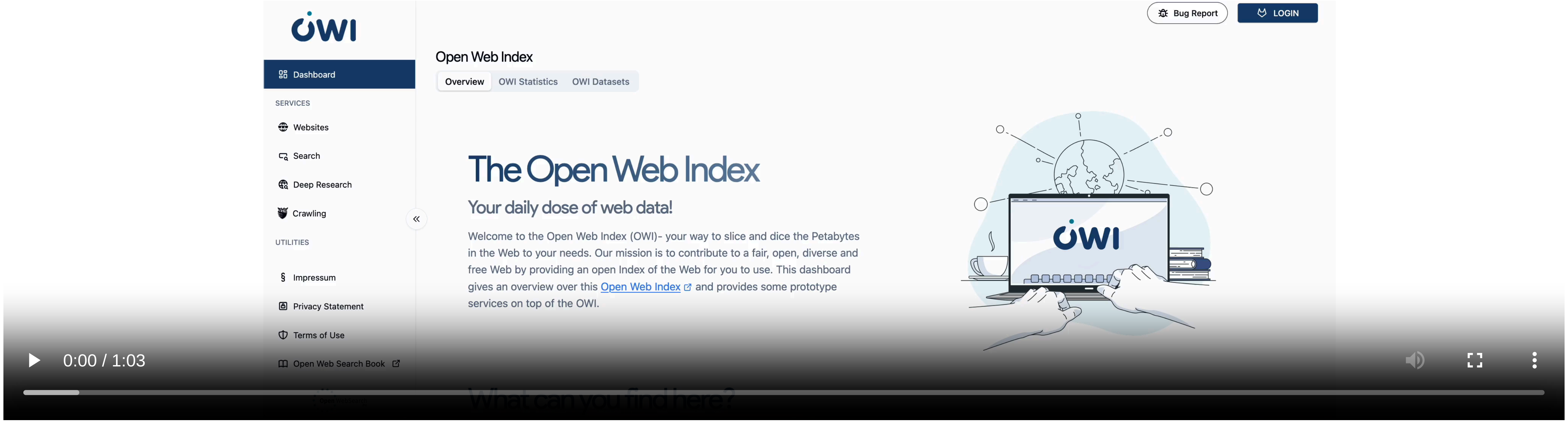
- Verified domain owners can submit **takedown requests** — subdomain or URL level
- Removed content is added to a blacklist; changelog propagated to all index users
- **Ethical self-assessment** survey required before enabling dataset downloads
- **OWLer crawler config**: express crawl constraints directly in the dashboard

subdomains

- **Domain verification**: prove ownership to unlock management capabilities

Transparency & Webmaster Engagement

- Dashboard provides full transparency on crawling activity
- Webmasters can claim ownership of their sites
- Express usage constraints and issue takedown requests
- Gen AI exclusion protocols: robots.txt, user-agent, X-Robots-Tag



FreeWebSearch Charter & Regulatory Reform

#FreeWebSearch Charter — launched 29 September 2025

10 Principles: transparency & explainability · competition & independent infrastructure · privacy without tracking · user/creator/advertiser control · equal access & non-discrimination · information diversity · environmental & social responsibility · integrity against disinformation · search competences & critical awareness · democratic control across borders

Four commitments: (a) serves society, people first; (b) respects democratic values; (c) transparent and open; (d) reflects internet diversity

81 signatories as of February 2026 — openwebsearch.eu/charter

3 Regulatory Reform Proposals

1. **Copyright** — new statutory exception for open web indexes in DSM Directive
2. **Data Protection** — public-interest exemption under Art. 6(1)(e) GDPR
3. **Interoperability** — classify OWI as neutral data intermediary under DSA

Promoted via EU Data Union Strategy consultation (July 2025) and European Parliament breakfast event (December 2025)

Societal Impact

- **Digital sovereignty** — reducing dependency on non-European search engines and AI data pipelines
- **Environmental sustainability** — reducing redundant crawling; shared infrastructure; data centre energy efficiency (CERN heat reuse)
- **Economic opportunity** — ecosystem enabling SMEs and startups; market potential study shows **ROI within 4 years**, projected **net benefit of ~4.5 B EUR over a decade**
- **Democracy & inclusiveness** — unbiased information access for informed citizens; multilingual/multicultural coverage; digital literacy

Proposal: European Web Data Clearing House

A public infrastructure that collects, preprocesses and serves bulk web data as a certified, legally compliant data source — eliminating liability risks for European innovators at scale

In Summary

What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

> OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

Key Messages

- Web-data and web search are **critical for Europe's digital sovereignty**
- **Reducing entry barriers** for researchers, innovators, and businesses
- Opening the core element in web search: the **Open Web Index**
- Open software stack, open pipelines, open data
- Distributed, data-centric, web-scale compute infrastructure
- Web data for **AI and Retrieval Augmented Generation**
- Building an **ecosystem** around data, software, and infrastructure
- Transparency and control for legal, ethical, and societal values

We are looking for ...

- **Researchers & tech innovators** — develop new search & retrieval paradigms and content analysis algorithms
- **Data centres** — help hosting a distributed Open Web Index
- **Industry & business partners** — discover the business models of an Open Web Index
- **Policy makers** — help shaping the governance of an open search ecosystem

Contact

- Email: join@openwebsearch.org
- Community: community@openwebsearch.eu
- Website: openwebsearch.eu
- Dashboard: openwebindex.eu
- Mattermost: openwebsearch.eu/community

Thanks. Questions? Comments? Feedback?

Before we come to the Questions

We are looking for ...

Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

Get in touch

Website	openwebsearch.eu
Dashboard	openwebindex.eu
Code Repository	https://opencode.it4i.eu/openwebsearcheu-public/
Community	openwebsearch.eu/community
Join	join@openwebsearch.org
General	community@openwebsearch.eu

Thank you — Questions, Comments, Feedback very welcome!